

# *Queimadas* Reference Satellite: A data analysis

Analysing data for choosing the next reference satellite for the Queimadas Program

Alber Sánchez<sup>1</sup>  
[alber.ipia@inpe.br](mailto:alber.ipia@inpe.br)

<sup>1</sup>National Institute for Space Research

May 27, 2026



# Agenda I

- 1 Introduction
- 2 Exploratory Data Analysis
- 3 Time Series
  - Time series decomposition
  - Reference satellites
  - Breakpoint identification
- 4 *Queimadas* method (12-month regression)
  - Regression
  - Forecast
- 5 *Queimadas* method (1-month regression)
  - Regression
  - Forecast
  - Forecast differences (12 versus 1 month)
- 6 Future work
- 7 References



# Introduction



# What's new?

- Queimadas method:
  - ▶ Outliers tables.
  - ▶ Confidence intervals.
  - ▶ Use of a month-by-month regression and forecast.
  - ▶ Forecasts' differences between Queimadas 12 vs. month-by-month.
  - ▶ BFast analysis.



# What would be the next reference satellite?

- What would be the next reference satellite [1, 3]?



# Online resources

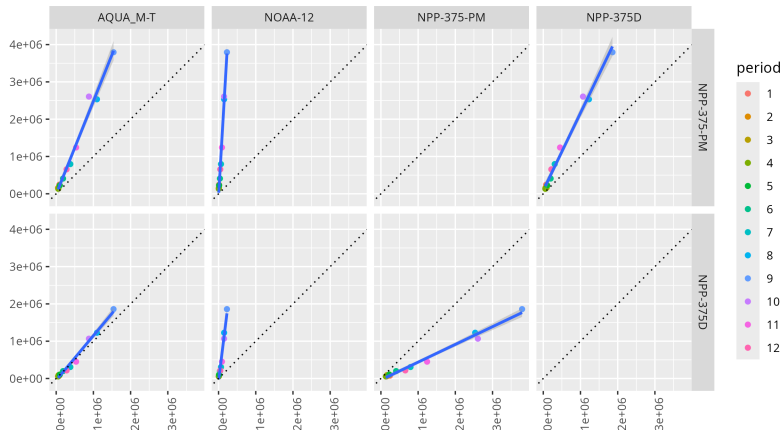
- Queimadas [portal](#).
- Queimadas [FAQ](#).
- Source code
  - ▶ Code repository available [here](#).
  - ▶ Slides available [here](#).



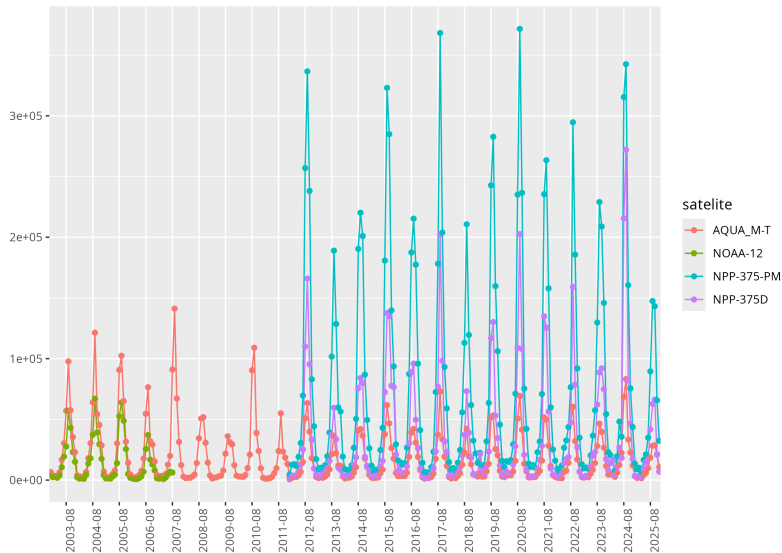
## Exploratory Data Analysis



# Brazilian fire foci aggregated by month (all years)

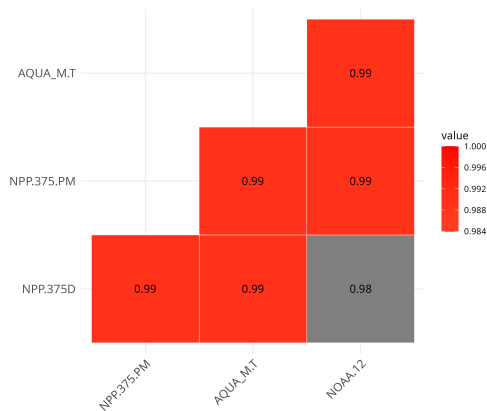


# Brazilian fire foci aggregated by year and month



# Correlation between fire foci from the reference satellites in Brazil

- The time series overlap partially.
- Use only the month (not the year) to compute correlation.
- This way we can use the whole time series in the estimation.

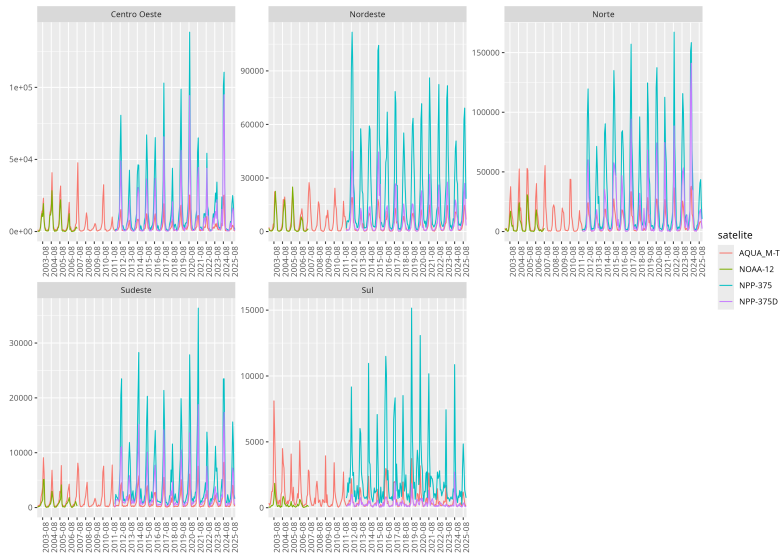


## Correlation between fire foci from the reference satellites in Brazil

- The time series overlap partially.
- Estimate the correlation using only observation that overlap.
- Some satellites don't overlap and their correlation can't be estimated.

| satelite_x | satelite_y | cor_val |
|------------|------------|---------|
| NOAA-12    | AQUA_M-T   | 0.922   |
| AQUA_M-T   | NPP-375-PM | 0.978   |
| AQUA_M-T   | NPP-375D   | 0.962   |
| NPP-375-PM | NPP-375D   | 0.958   |

# Brazilian regions by year and month



## Time Series

# Time series' components

- Structural time series model [2]:

- ▶ Trend  $g(t)$ .
- ▶ Periodic changes  $s(t)$ .
- ▶ Effects of holidays  $h(t)$ .

- **Assumptions:**

- ▶  $\epsilon_t$  is normally distributed.
- ▶ Forecasting is a curve-fitting problem; no account for temporal dependence structure in the data [4].

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

# Do the models fit?

- Do the data have a trend when smoothed?
- Do the data have a trend after applying the logarithm function and smoothing?
- Are the data autocorrelated in the time dimension?
- Do the data have a trend component?
- Do the data have season component?
- How does the model fit its observations?
- How do compare the residuals and the fitted values?
- Do the mode-observations residual have a normal distribution?



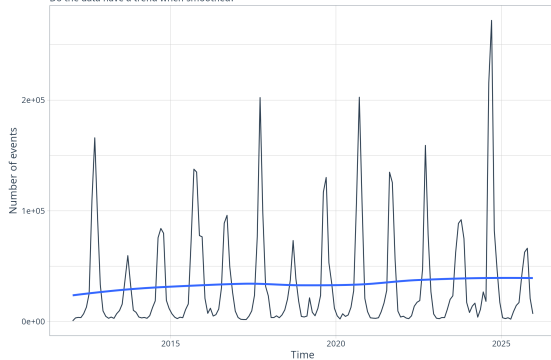
NPP-375D



# Time series trend (with smoothing)

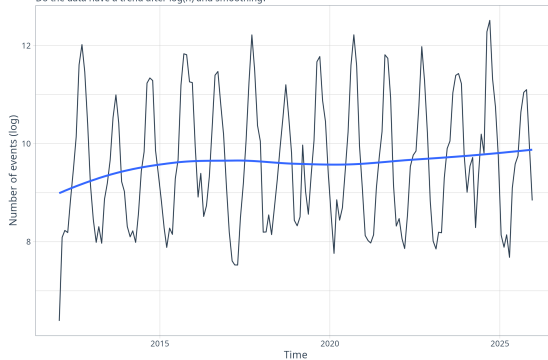
NPP-375D

Do the data have a trend when smoothed?

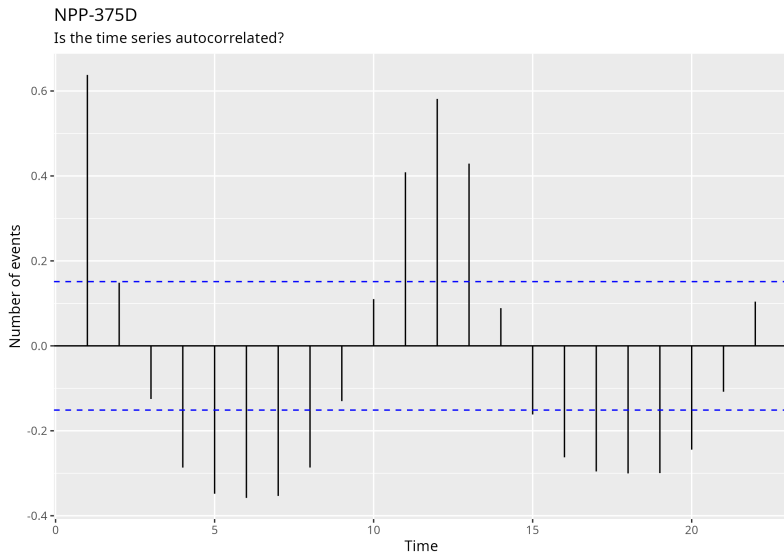


NPP-375D

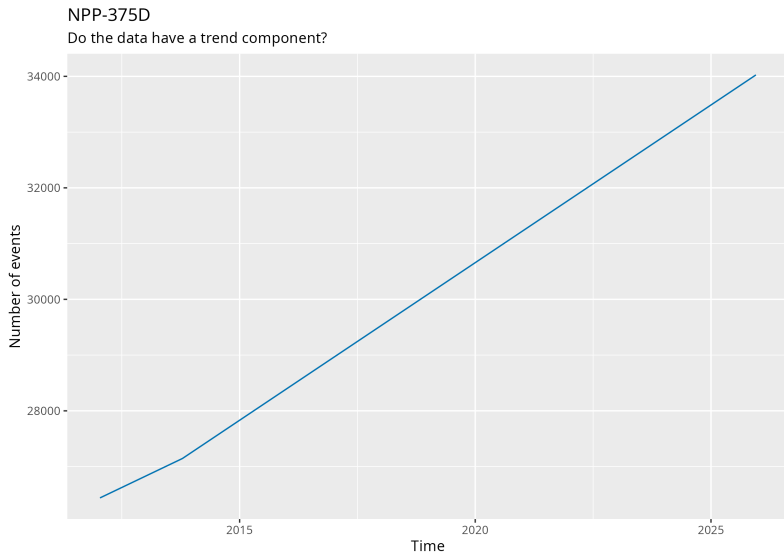
Do the data have a trend after log(n) and smoothing?



# Time series autocorrelation



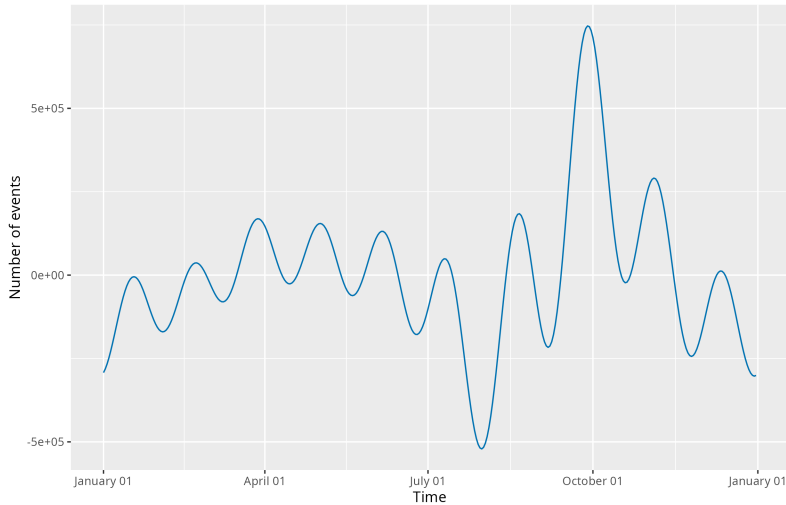
# Time series trend



# Time series seasonality

NPP-375D

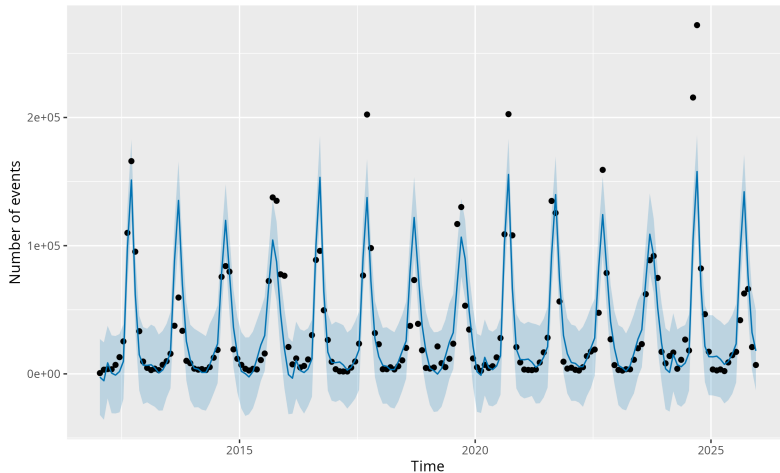
Do the data have a seasonality component?



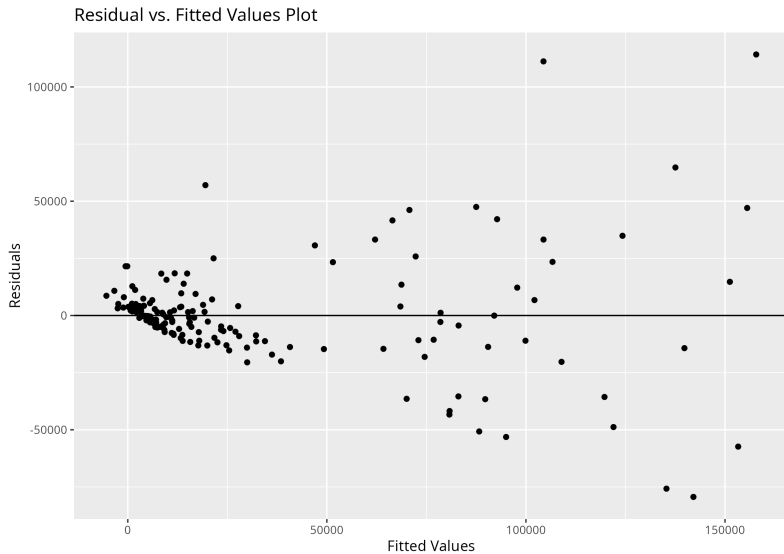
# Time series versus forecast

NPP-375D

Observations (points) versus forecasting (line)



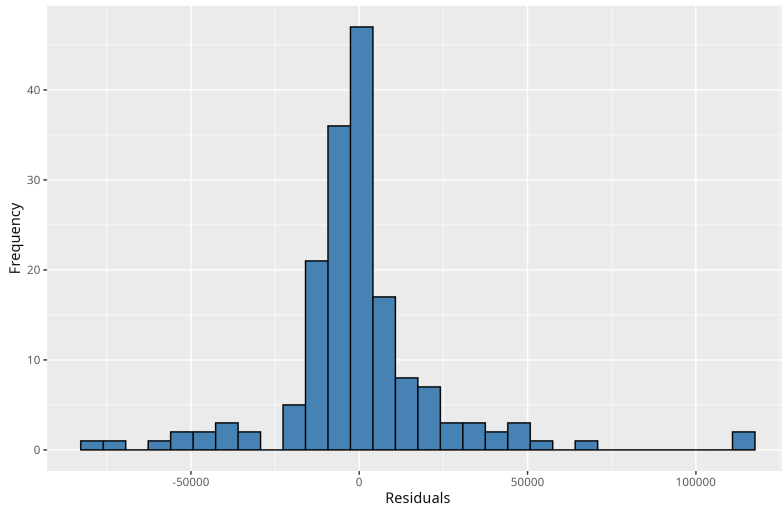
# Time series versus forecast



# Time series versus forecast

## Histogram of Residuals

Do the residuals have a normal distribution?



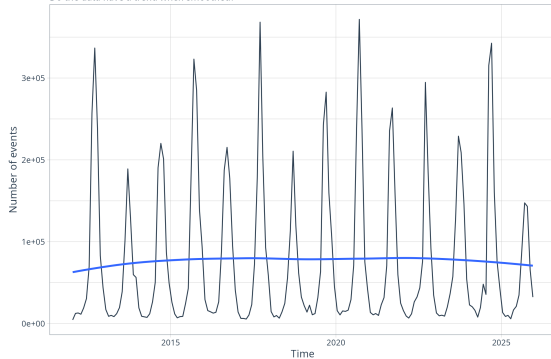
NPP-375-PM



# Time series trend (with smoothing)

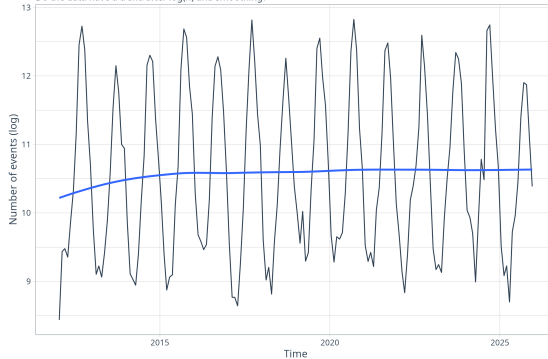
NPP-375-PM

Do the data have a trend when smoothed?



NPP-375-PM

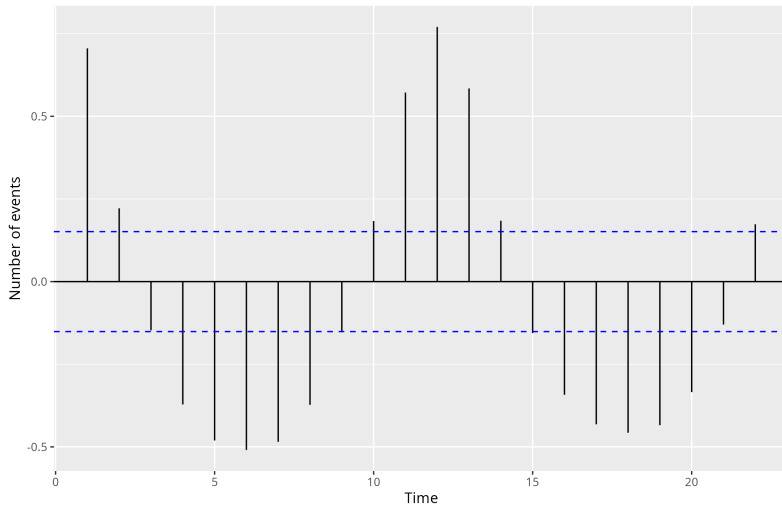
Do the data have a trend after log(n) and smoothing?



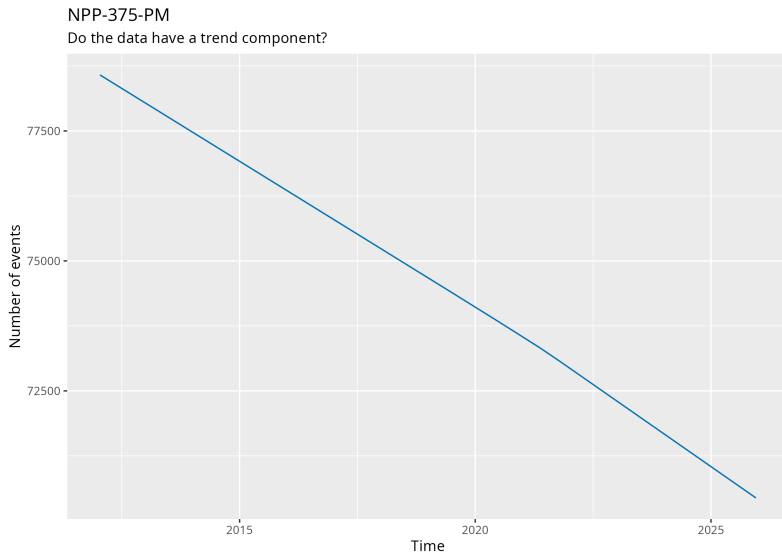
# Time series autocorrelation

NPP-375-PM

Is the time series autocorrelated?



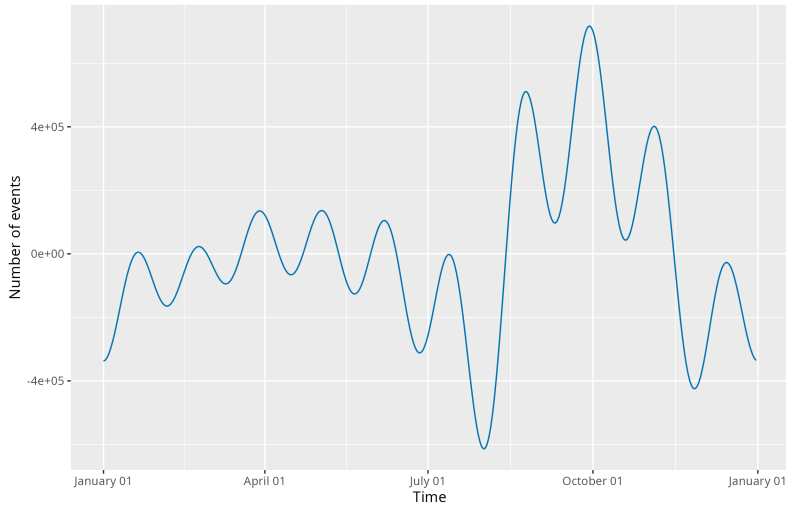
# Time series trend



# Time series seasonality

NPP-375-PM

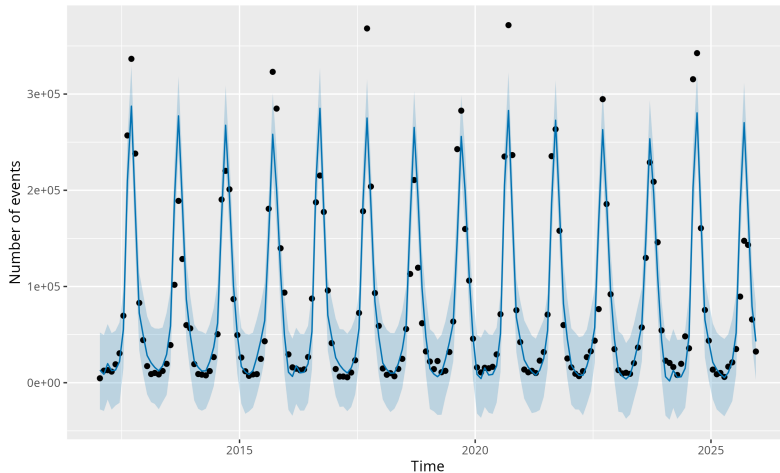
Do the data have a seasonality component?



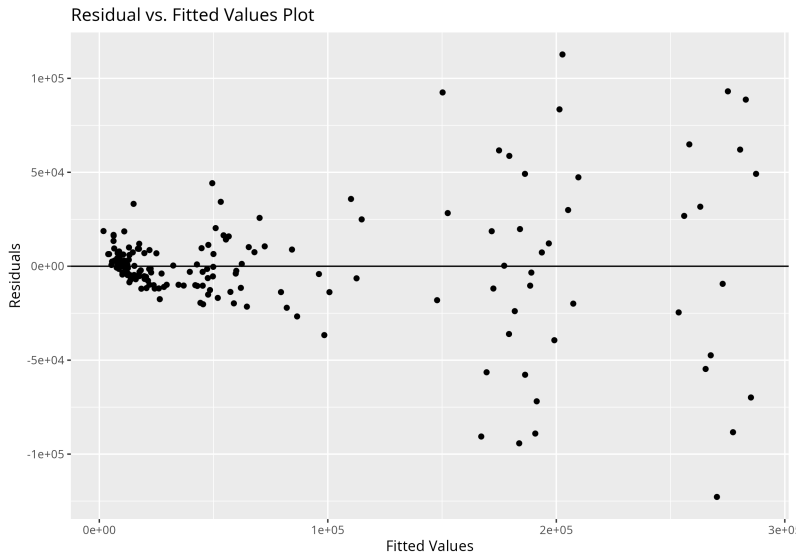
# Time series versus forecast

NPP-375-PM

Observations (points) versus forecasting (line)



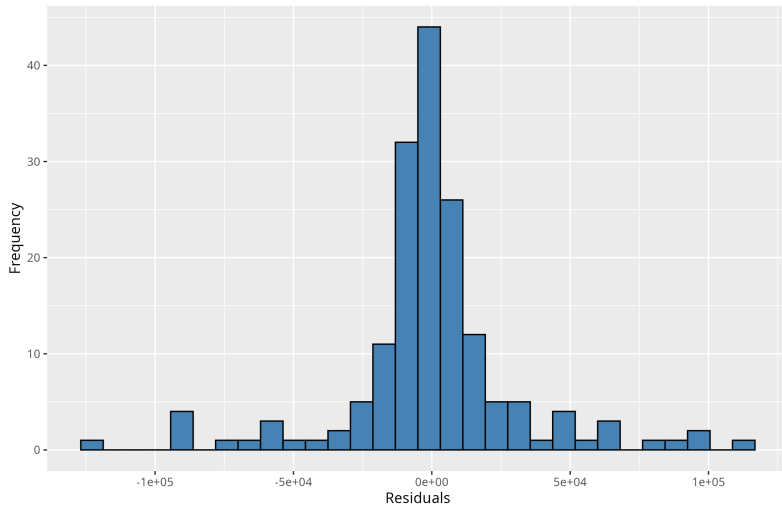
# Time series versus forecast



# Time series versus forecast

## Histogram of Residuals

Do the residuals have a normal distribution?



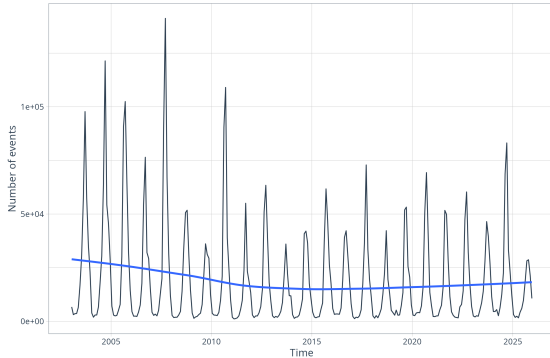
AQUA M T



# Time series trend (with smoothing)

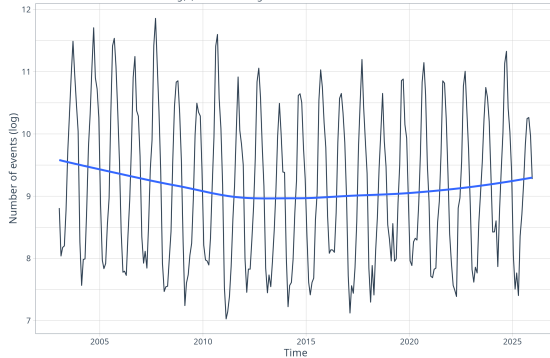
AQUA\_M-T

Do the data have a trend when smoothed?

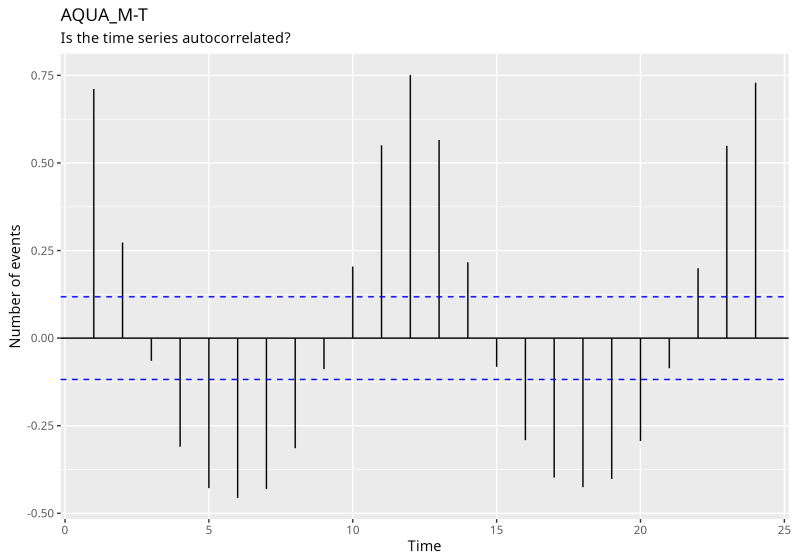


AQUA\_M-T

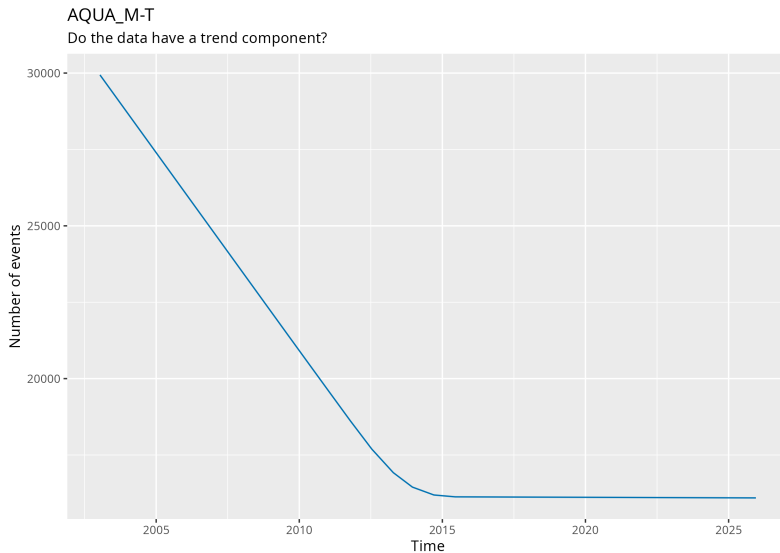
Do the data have a trend after log(n) and smoothing?



# Time series autocorrelation



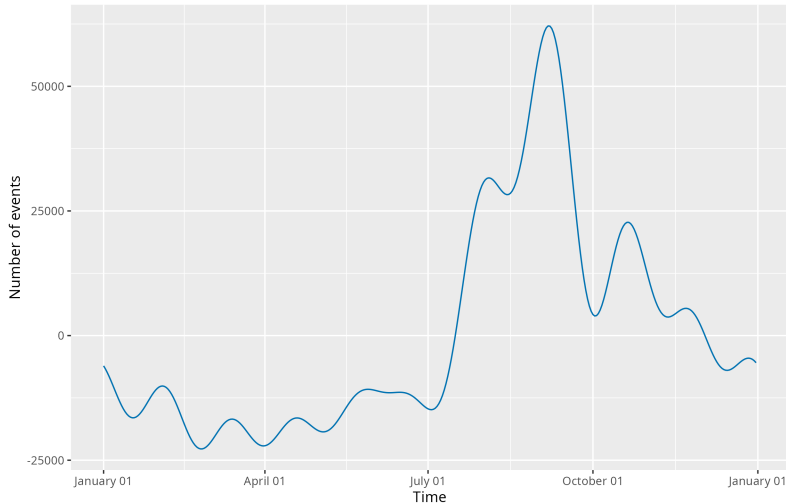
# Time series trend



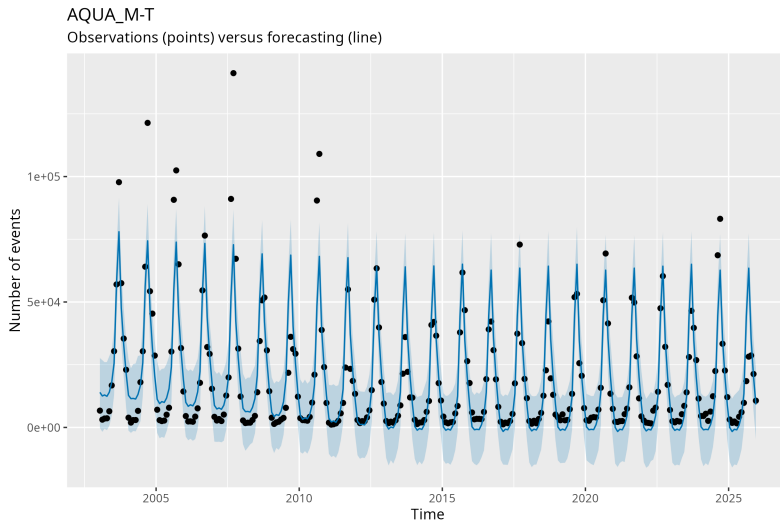
# Time series seasonality

AQUA\_M-T

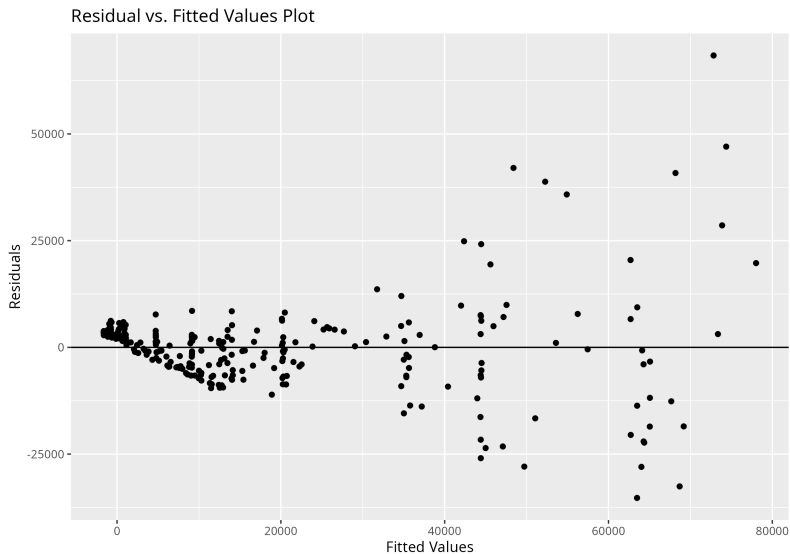
Do the data have a seasonality component?



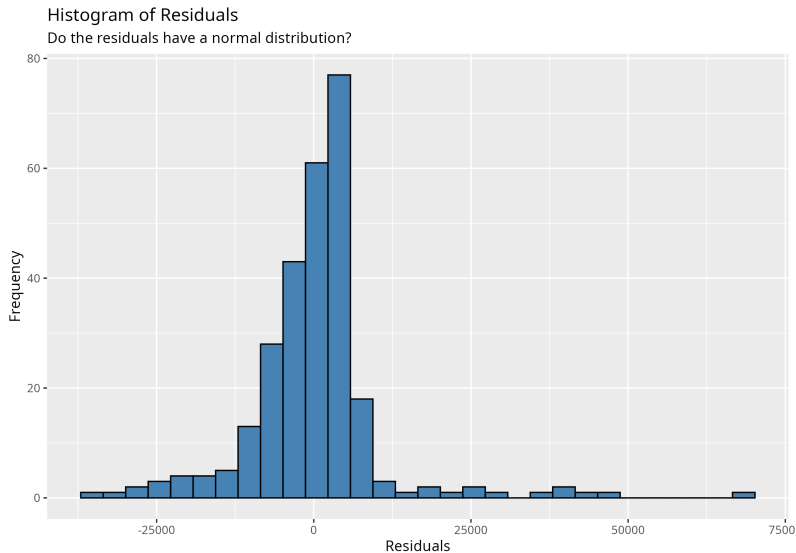
# Time series versus forecast



# Time series versus forecast



# Time series versus forecast

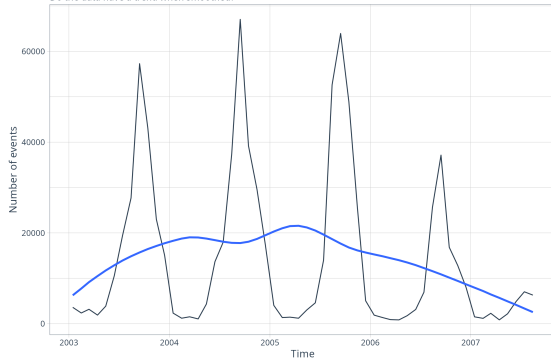


NOAA 12

# Time series trend (with smoothing)

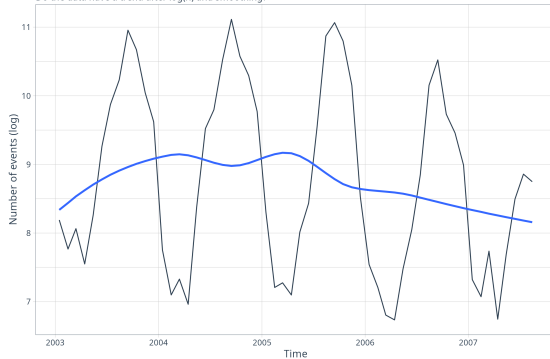
NOAA-12

Do the data have a trend when smoothed?



NOAA-12

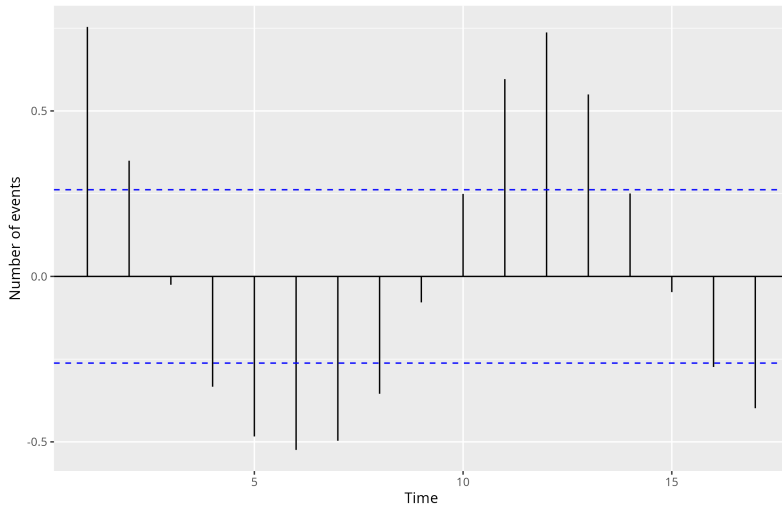
Do the data have a trend after log(n) and smoothing?



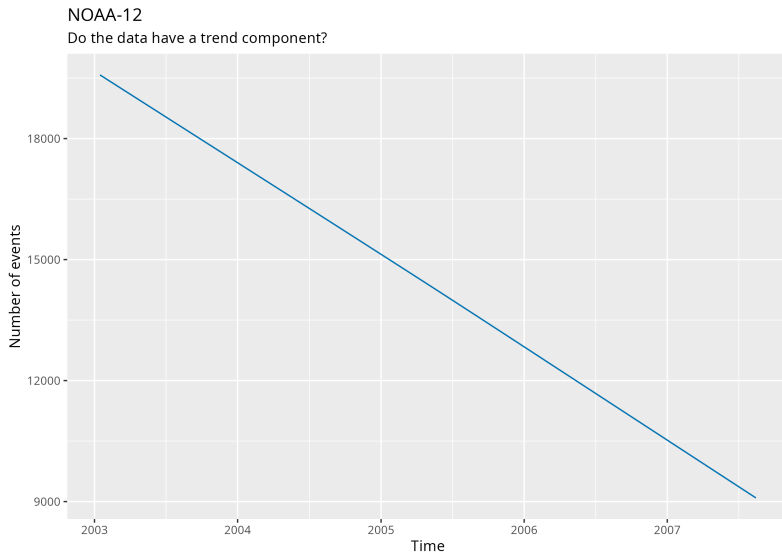
# Time series autocorrelation

NOAA-12

Is the time series autocorrelated?



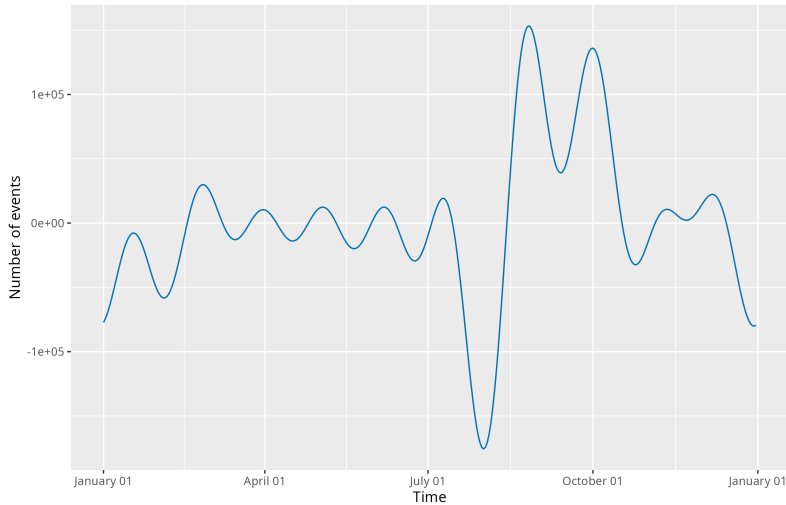
# Time series trend



# Time series seasonality

NOAA-12

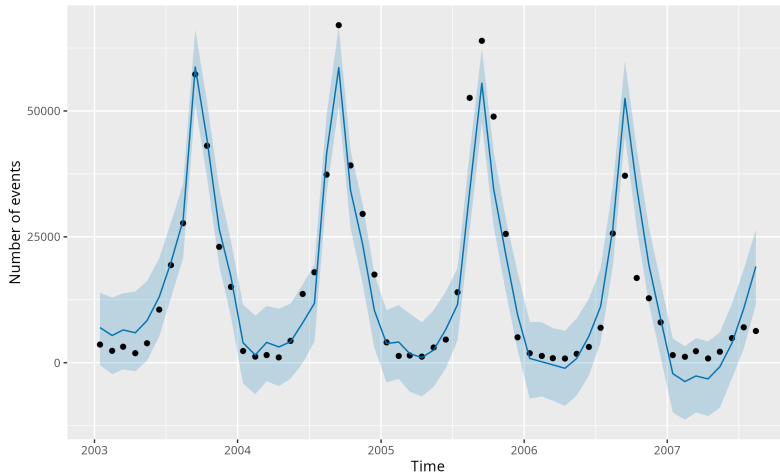
Do the data have a seasonality component?



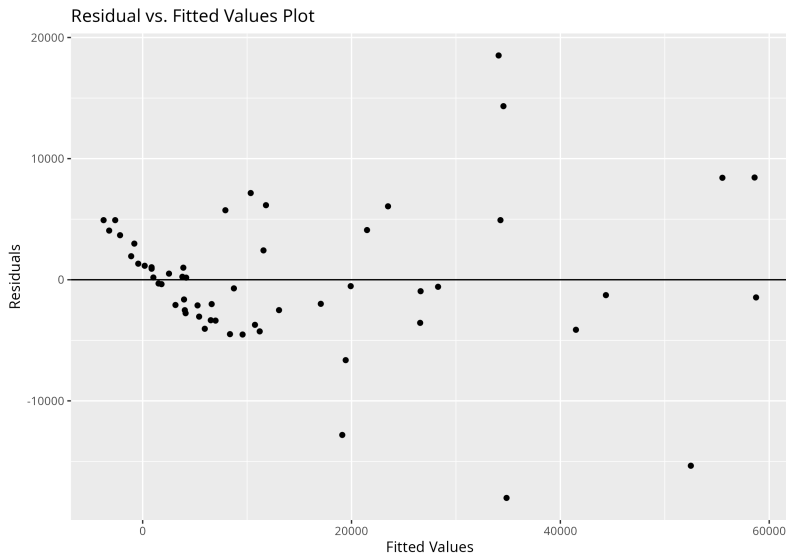
# Time series versus forecast

NOAA-12

Observations (points) versus forecasting (line)



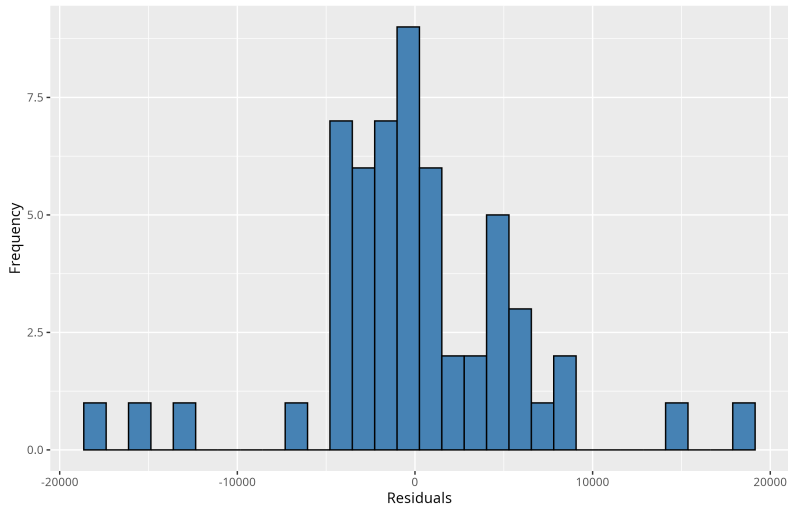
# Time series versus forecast



# Time series versus forecast

Histogram of Residuals

Do the residuals have a normal distribution?



# Summary

| Name       | Trend | TLog | Autocor | Season | Random Res | Res Dist |
|------------|-------|------|---------|--------|------------|----------|
| NPP-375D   | +     | +    | 12 m    | yes    | TODO       | Normal   |
| NPP-375-PM | +     | +    | 12 m    | yes    | TODO       | Normal   |
| AQUA M T   | -+    | -+   | 12 m    | yes    | TODO       | Normal   |
| NOAA 12    | -     | +-   | 12 m    | yes    | TODO       | Normal   |

BFast

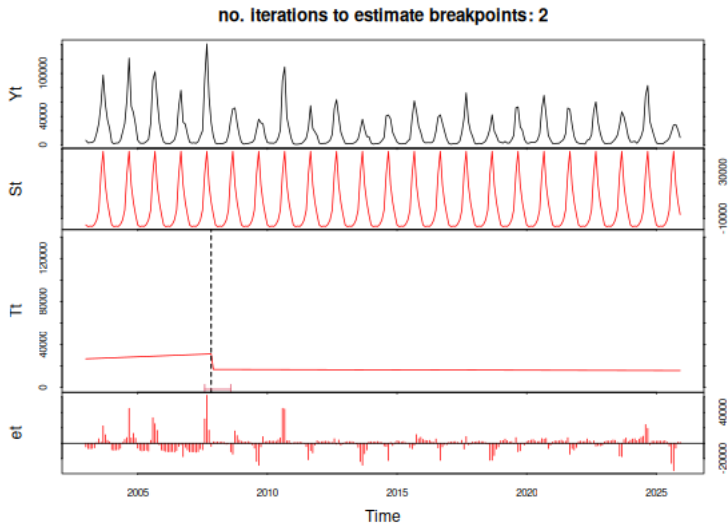


# Breaks For Additive Seasonal and Trend - BFAST

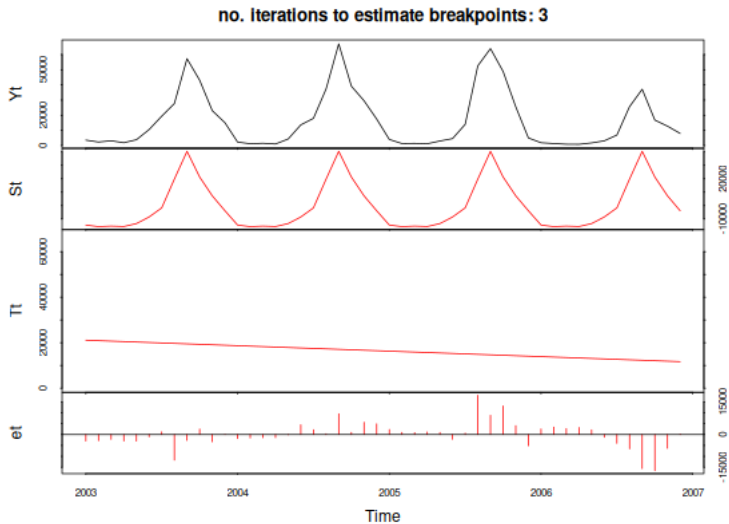
- Additive decomposition model that iteratively fits a linear trend and a seasonal model [5, 6].
- $Y_t$  is the observed data at time  $t$ .
- $T_t$  is the trend component.
- $S_t$  is the season component.
- $e_t$  is the remaining variation in the data beyond the seasonal and trend components.

$$Y_t = T_t + S_t + e_t$$

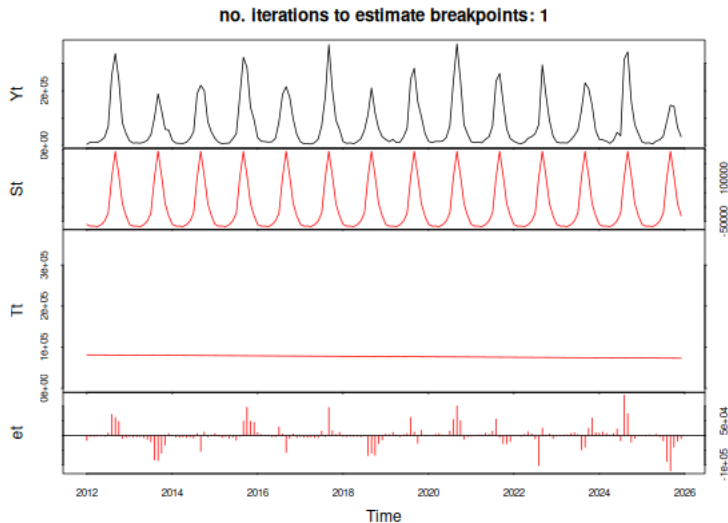
# Time series Breaks (BFast) AQUA M-T



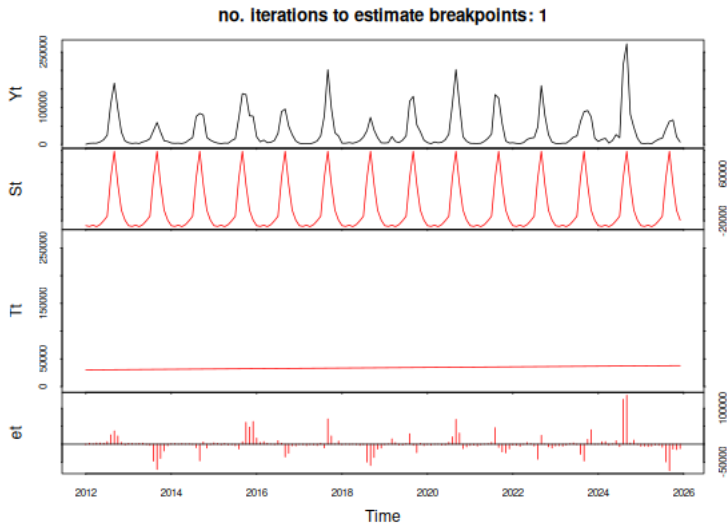
# Time series Breaks (BFast) NOAA-12



# Time series Breaks (BFast) NPP-375-PM



# Time series Breaks (BFast) NPP-375-PM



*Queimadas* method (12-month regression)

## Queimadas method for forecasting heat foci

- 1 Get time series of two reference satellites:
  - ▶ One which we would like to extend in time (A.K.A.  $y$ ).
  - ▶ The other one covering the whole time line we would like to forecast (A.K.A.  $x$ ).
- 2 Adjust a linear model ( $y = Ax + b$ ) to the overlapping observations in the time series of two reference satellites, using only the month of the observations.
- 3 Use the model to predict the whole time line.
- 4 Check forecast results.

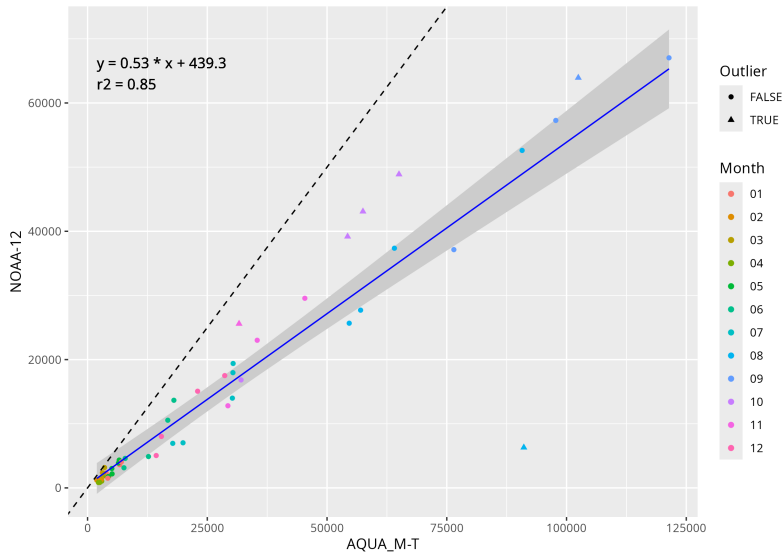


# Outliers and confidence intervals

- Outliers were determined on the regression residuals.
- The outliers are outside the range given by  $\pm 1.58 IQR / \sqrt{n}$ , where IQR is the Interquartile range, that is, the range between the first and third quartiles.
- The confidence intervals (gray shadows in the regression and forecast figures) is given the 95% confidence level.



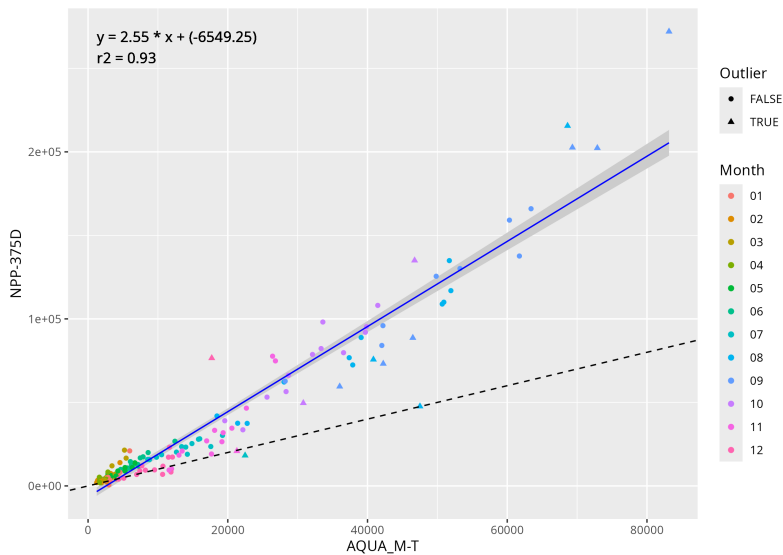
# Linear model NOAA-12 vs. AQUA\_M-T



## Outliers linear model NOAA-12 vs. AQUA\_M-T

| Year | Month | x      | y     | yhat     | Residual |
|------|-------|--------|-------|----------|----------|
| 2003 | 10    | 57495  | 43089 | 31166.34 | 28490.85 |
| 2004 | 10    | 54292  | 39184 | 29454.56 | 26916.05 |
| 2005 | 09    | 102455 | 63933 | 55194.30 | 50159.10 |
| 2005 | 10    | 65023  | 48878 | 35189.53 | 32166.01 |
| 2005 | 11    | 31631  | 25587 | 17343.85 | 15445.88 |
| 2007 | 08    | 91085  | 6299  | 49117.84 | 44723.41 |

# Linear model NPP-375D vs. AQUA\_M-T

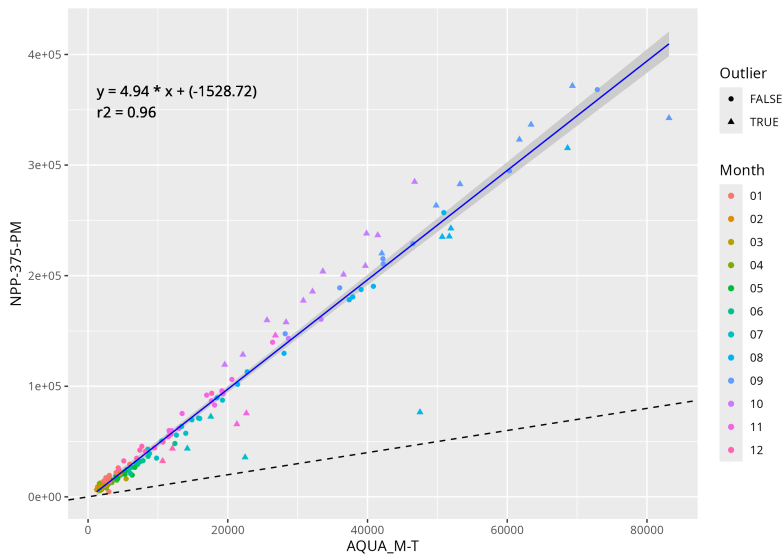


## Outliers linear model NPP-375D vs. AQUA\_M-T

| Year | Month | x     | y      | yhat      | Residual  |
|------|-------|-------|--------|-----------|-----------|
| 2013 | 09    | 36019 | 59535  | 85270.83  | 82298.38  |
| 2014 | 08    | 40845 | 75686  | 97573.33  | 94182.70  |
| 2015 | 10    | 46741 | 135011 | 112603.49 | 108663.82 |
| 2015 | 12    | 17703 | 76526  | 38579.46  | 36598.43  |
| 2016 | 10    | 30809 | 49627  | 71989.44  | 69419.85  |
| 2017 | 09    | 72895 | 202308 | 179275.59 | 172670.03 |
| 2018 | 09    | 42251 | 73180  | 101157.53 | 97639.17  |
| 2020 | 09    | 69329 | 202617 | 170185.10 | 163954.96 |
| 2022 | 08    | 47507 | 47640  | 114556.19 | 110542.91 |
| 2023 | 09    | 46486 | 88659  | 111953.44 | 108038.17 |
| 2024 | 07    | 22478 | 18231  | 50751.95  | 48651.14  |
| 2024 | 08    | 68635 | 215589 | 168415.95 | 162258.60 |
| 2024 | 09    | 83154 | 272041 | 205427.96 | 197731.67 |
| 2025 | 11    | 21301 | 20907  | 47751.53  | 45692.14  |



# Linear model NPP-375-PM vs. AQUA\_M-T

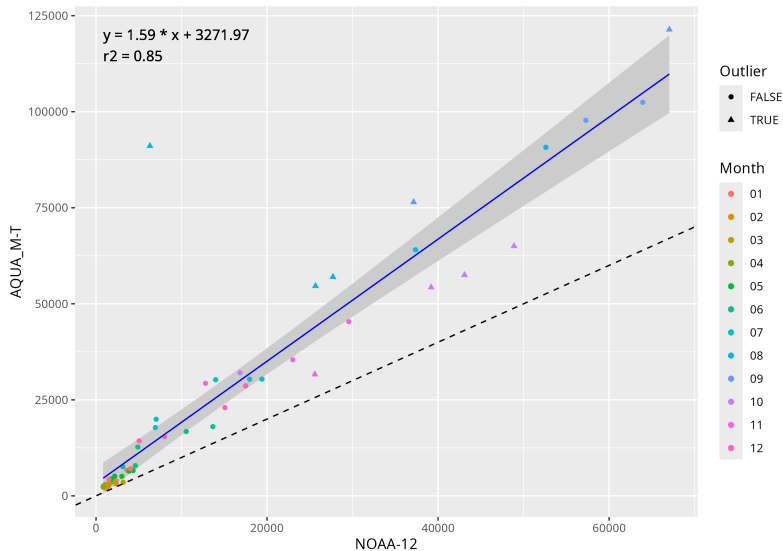


# Outliers linear model NPP-375-PM vs. AQUA\_M-T I

| Year | Month | x     | y      | yhat      | Residual  |
|------|-------|-------|--------|-----------|-----------|
| 2012 | 09    | 63408 | 336595 | 311987.67 | 303878.61 |
| 2012 | 10    | 39860 | 238154 | 195556.24 | 190784.66 |
| 2013 | 10    | 22159 | 128579 | 108034.89 | 105016.89 |
| 2014 | 09    | 42049 | 220178 | 206379.60 | 201322.96 |
| 2014 | 10    | 36568 | 201003 | 179279.18 | 174918.49 |
| 2015 | 09    | 61739 | 323070 | 303735.42 | 295875.58 |
| 2015 | 10    | 46741 | 284887 | 229578.86 | 223886.79 |
| 2016 | 10    | 30809 | 177465 | 150804.21 | 147091.65 |
| 2017 | 07    | 17568 | 72520  | 85335.02  | 82474.89  |
| 2017 | 10    | 33607 | 203876 | 164638.72 | 160624.54 |
| 2018 | 10    | 19568 | 119528 | 95223.87  | 92316.61  |
| 2019 | 08    | 51935 | 242744 | 255260.23 | 248835.87 |
| 2019 | 09    | 53234 | 282749 | 261683.04 | 255071.92 |
| 2019 | 10    | 25613 | 159766 | 125112.95 | 121870.04 |
| 2020 | 08    | 50694 | 235086 | 249124.19 | 242877.03 |
| 2020 | 09    | 69329 | 371618 | 341263.64 | 332262.29 |
| 2020 | 10    | 41468 | 236591 | 203506.88 | 198526.71 |
| 2021 | 08    | 51711 | 235453 | 254152.68 | 247760.40 |
| 2021 | 09    | 49829 | 263437 | 244847.26 | 238722.83 |
| 2021 | 10    | 28342 | 157917 | 138606.30 | 135133.53 |
| 2022 | 07    | 14212 | 43692  | 68741.51  | 65880.85  |
| 2022 | 08    | 47507 | 76511  | 233366.29 | 227567.89 |
| 2022 | 10    | 32130 | 185691 | 157335.80 | 153484.36 |
| 2023 | 10    | 39692 | 208865 | 194725.58 | 189975.52 |
| 2023 | 11    | 26819 | 145959 | 131075.93 | 127736.52 |
| 2024 | 07    | 22478 | 35755  | 109612.17 | 106576.90 |
| 2024 | 08    | 68635 | 315415 | 337832.20 | 328936.02 |
| 2024 | 09    | 83154 | 342525 | 409620.38 | 398500.72 |
| 2024 | 11    | 22663 | 75577  | 110526.88 | 107481.26 |



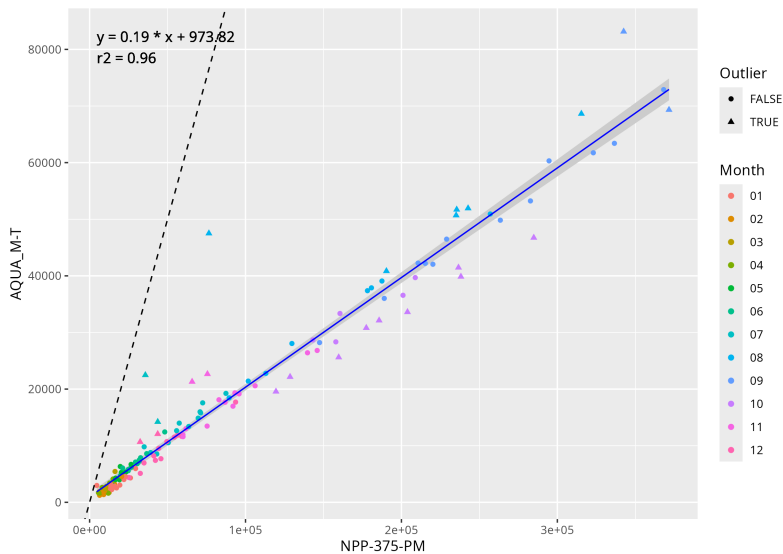
# Linear model AQUA\_M-T vs. NOAA-12



## Outliers linear model AQUA\_M-T vs. NOAA-12

| Year | Month | x     | y      | yhat      | Residual  |
|------|-------|-------|--------|-----------|-----------|
| 2003 | 08    | 27705 | 57004  | 47299.24  | 43274.404 |
| 2003 | 10    | 43089 | 57495  | 71746.65  | 65612.335 |
| 2004 | 09    | 67038 | 121395 | 109805.09 | 99685.055 |
| 2004 | 10    | 39184 | 54292  | 65541.04  | 60000.759 |
| 2005 | 10    | 48878 | 65023  | 80946.21  | 73890.511 |
| 2005 | 11    | 25587 | 31631  | 43933.43  | 40127.458 |
| 2006 | 08    | 25670 | 54630  | 44065.33  | 40251.283 |
| 2006 | 09    | 37137 | 76475  | 62288.06  | 57047.282 |
| 2007 | 08    | 6299  | 91085  | 13282.00  | 9725.217  |

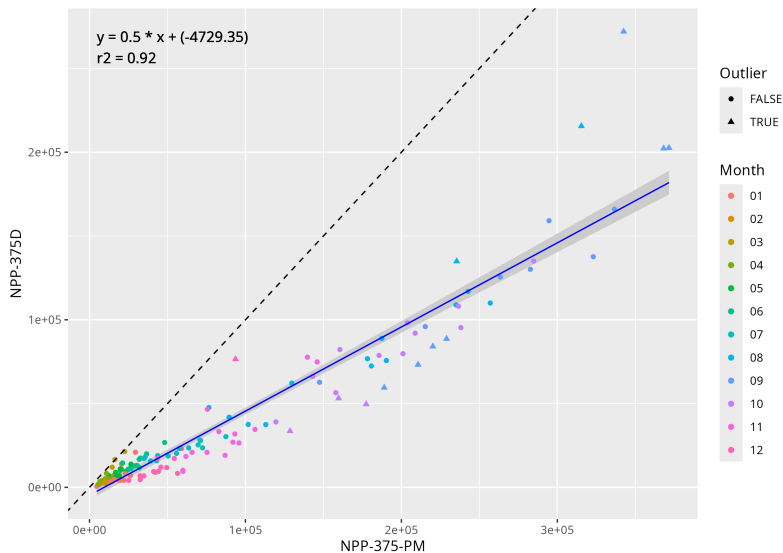
# Linear model AQUA\_M-T vs. NPP-375-PM



# Outliers linear model AQUA\_M-T vs. NPP-375-PM

| Year | Month | x      | y     | yhat      | Residual  |
|------|-------|--------|-------|-----------|-----------|
| 2012 | 10    | 238154 | 39860 | 47087.642 | 45932.821 |
| 2013 | 10    | 128579 | 22159 | 25870.605 | 25221.508 |
| 2014 | 08    | 190370 | 40845 | 37835.213 | 36929.628 |
| 2015 | 10    | 284887 | 46741 | 56136.565 | 54719.203 |
| 2016 | 10    | 177465 | 30809 | 35336.415 | 34492.563 |
| 2017 | 10    | 203876 | 33607 | 40450.384 | 39477.197 |
| 2018 | 10    | 119528 | 19568 | 24118.058 | 23495.127 |
| 2019 | 08    | 242744 | 51935 | 47976.405 | 46796.422 |
| 2019 | 10    | 159766 | 25613 | 31909.352 | 31144.376 |
| 2020 | 08    | 235086 | 50694 | 46493.584 | 45355.487 |
| 2020 | 09    | 371618 | 69329 | 72930.312 | 71002.679 |
| 2020 | 10    | 236591 | 41468 | 46784.998 | 45638.707 |
| 2021 | 08    | 235453 | 51711 | 46564.647 | 45424.553 |
| 2022 | 07    | 43692  | 14212 | 9433.912  | 8832.751  |
| 2022 | 08    | 76511  | 47507 | 15788.664 | 15225.096 |
| 2022 | 10    | 185691 | 32130 | 36929.217 | 36046.370 |
| 2024 | 07    | 35755  | 22478 | 7897.069  | 7276.859  |
| 2024 | 08    | 315415 | 68635 | 62047.710 | 60453.197 |
| 2024 | 09    | 342525 | 83154 | 67297.027 | 65542.741 |
| 2024 | 11    | 75577  | 22663 | 15607.814 | 15044.175 |
| 2024 | 12    | 43682  | 12075 | 9431.976  | 8830.792  |
| 2025 | 11    | 65739  | 21301 | 13702.879 | 13134.834 |
| 2025 | 12    | 32408  | 10669 | 7248.988  | 6619.743  |

# Linear model NPP-375D vs. NPP-375-PM

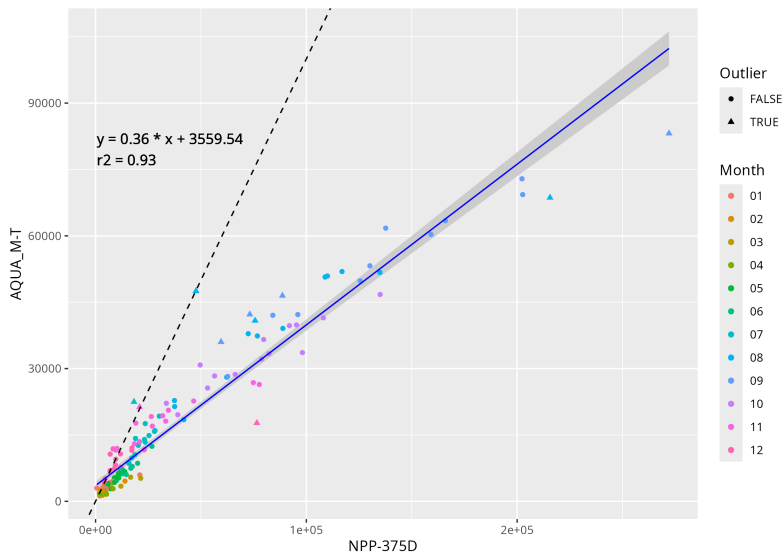


## Outliers linear model NPP-375D vs. NPP-375-PM

| Year | Month | x      | y      | yhat      | Residual  |
|------|-------|--------|--------|-----------|-----------|
| 2013 | 09    | 188981 | 59535  | 90176.05  | 86860.92  |
| 2013 | 10    | 128579 | 33591  | 59842.44  | 57448.30  |
| 2014 | 09    | 220178 | 84087  | 105843.04 | 101940.58 |
| 2015 | 12    | 93633  | 76526  | 42292.72  | 40179.44  |
| 2016 | 10    | 177465 | 49627  | 84392.77  | 81280.29  |
| 2017 | 09    | 368188 | 202308 | 180172.99 | 173138.78 |
| 2018 | 09    | 210659 | 73180  | 101062.64 | 97344.35  |
| 2019 | 10    | 159766 | 53159  | 75504.41  | 72682.86  |
| 2020 | 09    | 371618 | 202617 | 181895.52 | 174785.62 |
| 2021 | 08    | 235453 | 134894 | 113514.08 | 109308.94 |
| 2023 | 09    | 228974 | 88659  | 110260.35 | 106184.58 |
| 2024 | 08    | 315415 | 215589 | 153670.63 | 147789.41 |
| 2024 | 09    | 342525 | 272041 | 167285.15 | 160814.62 |



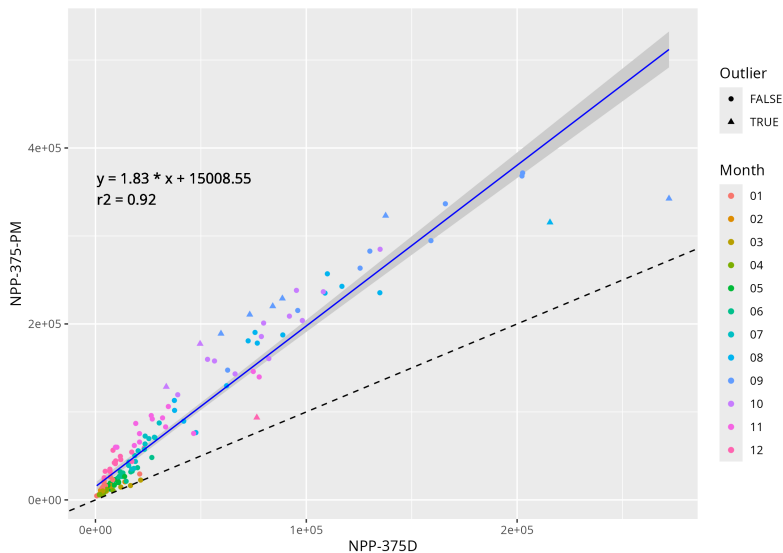
# Linear model AQUA\_M-T vs. NPP-375D



## Outliers linear model AQUA\_M-T vs. NPP-375D

| Year | Month | x      | y     | yhat      | Residual  |
|------|-------|--------|-------|-----------|-----------|
| 2013 | 09    | 59535  | 36019 | 25179.73  | 24333.935 |
| 2014 | 08    | 75686  | 40845 | 31044.98  | 30052.502 |
| 2015 | 12    | 76526  | 17703 | 31350.03  | 30348.737 |
| 2018 | 09    | 73180  | 42251 | 30134.92  | 29168.135 |
| 2022 | 08    | 47640  | 47507 | 20860.05  | 20085.684 |
| 2023 | 09    | 88659  | 46486 | 35756.14  | 34617.960 |
| 2024 | 07    | 18231  | 22478 | 10180.14  | 9395.772  |
| 2024 | 08    | 215589 | 68635 | 81850.88  | 78893.926 |
| 2024 | 09    | 272041 | 83154 | 102351.48 | 98526.885 |
| 2025 | 11    | 20907  | 21301 | 11151.93  | 10379.892 |

# Linear model NPP-375-PM vs. NPP-375D



## Outliers linear model NPP-375-PM vs. NPP-375D

| Year | Month | x      | y      | yhat      | Residual  |
|------|-------|--------|--------|-----------|-----------|
| 2013 | 09    | 59535  | 188981 | 123768.55 | 119261.28 |
| 2013 | 10    | 33591  | 128579 | 76373.41  | 72408.77  |
| 2014 | 09    | 84087  | 220178 | 168620.75 | 162840.52 |
| 2015 | 09    | 137635 | 323070 | 266443.56 | 256879.43 |
| 2015 | 12    | 76526  | 93633  | 154808.13 | 149472.22 |
| 2016 | 10    | 49627  | 177465 | 105668.37 | 101492.36 |
| 2018 | 09    | 73180  | 210659 | 148695.58 | 143543.52 |
| 2023 | 09    | 88659  | 228974 | 176973.00 | 170907.60 |
| 2024 | 08    | 215589 | 315415 | 408851.85 | 393094.09 |
| 2024 | 09    | 272041 | 342525 | 511979.76 | 491598.32 |

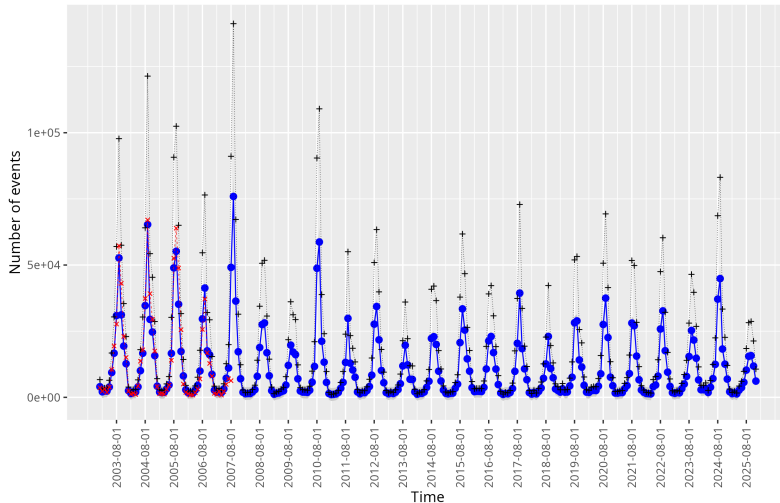
## Linear regression parameters

| satellite_x | satellite_y | equation                      | r2   |
|-------------|-------------|-------------------------------|------|
| AQUA_M-T    | NOAA-12     | $y = 0.534 * x + 439.303$     | 0.85 |
| AQUA_M-T    | NPP-375-PM  | $y = 4.944 * x + (-1528.724)$ | 0.96 |
| AQUA_M-T    | NPP-375D    | $y = 2.549 * x + (-6549.248)$ | 0.93 |
| NOAA-12     | AQUA_M-T    | $y = 1.589 * x + 3271.974$    | 0.85 |
| NPP-375-PM  | AQUA_M-T    | $y = 0.194 * x + 973.819$     | 0.96 |
| NPP-375-PM  | NPP-375D    | $y = 0.502 * x + (-4729.351)$ | 0.92 |
| NPP-375D    | AQUA_M-T    | $y = 0.363 * x + 3559.538$    | 0.93 |
| NPP-375D    | NPP-375-PM  | $y = 1.827 * x + 15008.547$   | 0.92 |

# Stretch NOAA-12 to match AQUA\_M-T

Transformation of AQUA\_M-T into NOAA-12

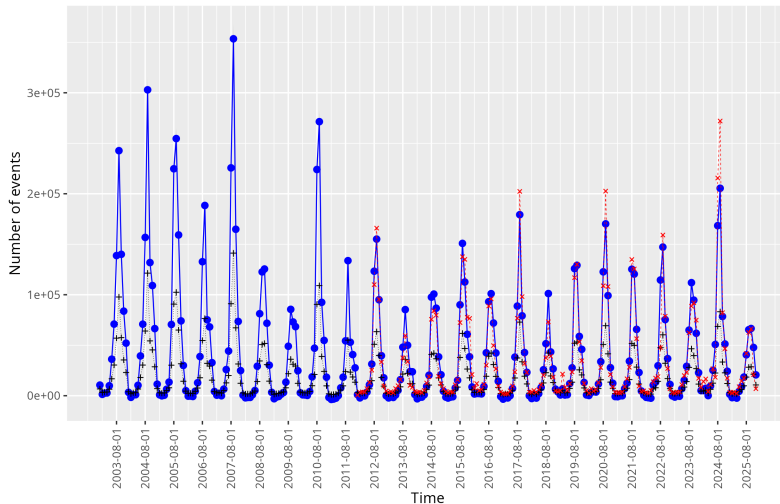
Forecast (blue line) versus observations (NOAA-12 red crosses, AQUA\_M-T black plusses)



# Stretch NPP-375D to match AQUA\_M-T

Transformation of AQUA\_M-T into NPP-375D

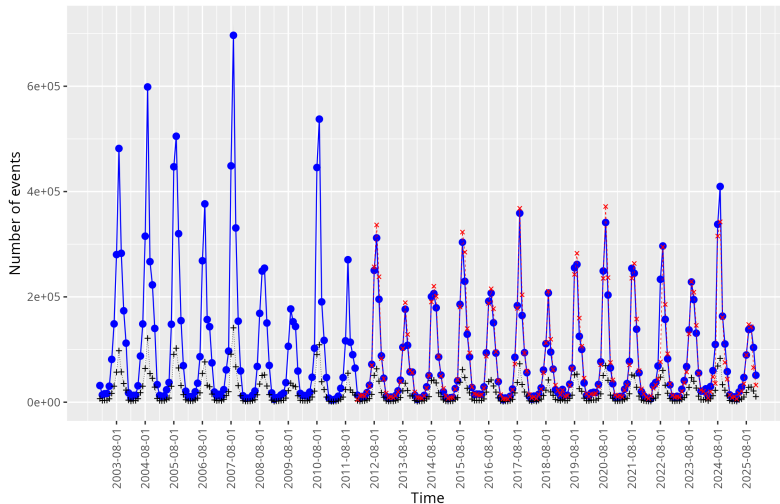
Forecast (blue line) versus observations (NPP-375D red crosses, AQUA\_M-T black plusses)



# Stretch NPP-375-PM to match AQUA\_M-T

Transformation of AQUA\_M-T into NPP-375-PM

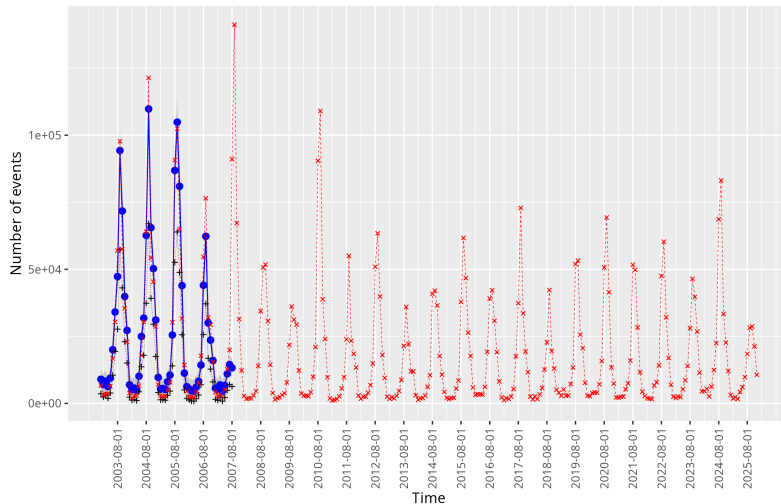
Forecast (blue line) versus observations (NPP-375-PM red crosses, AQUA\_M-T black plusses)



# Stretch match AQUA\_M-T to match NOAA-12

Transformation of NOAA-12 into AQUA\_M-T

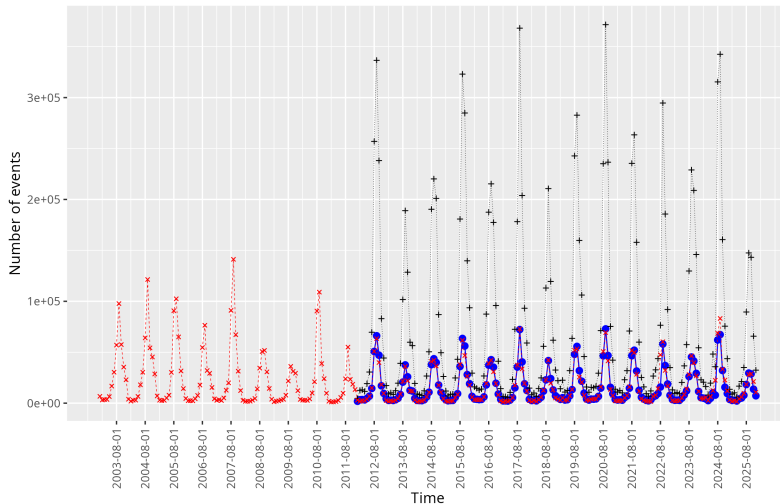
Forecast (blue line) versus observations (AQUA\_M-T red crosses, NOAA-12 black plusses)



# Stretch match AQUA\_M-T to match NPP-375-PM

Transformation of NPP-375-PM into AQUA\_M-T

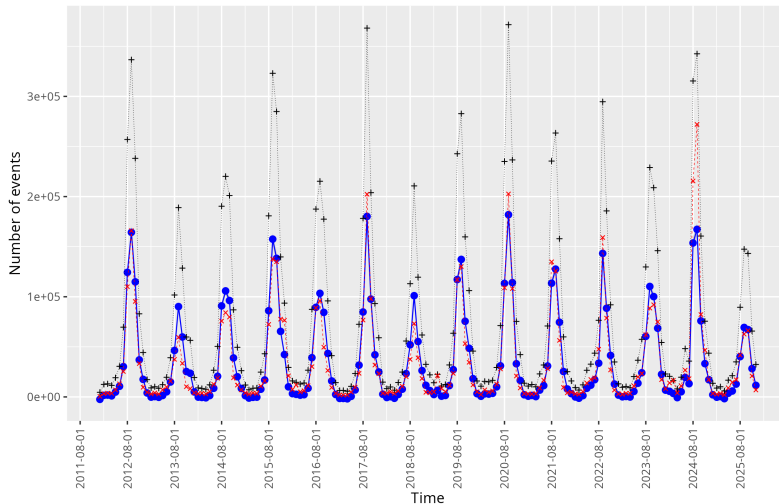
Forecast (blue line) versus observations (AQUA\_M-T red crosses, NPP-375-PM black plusses)



# Stretch match NPP-375D to match NPP-375-PM

Transformation of NPP-375-PM into NPP-375D

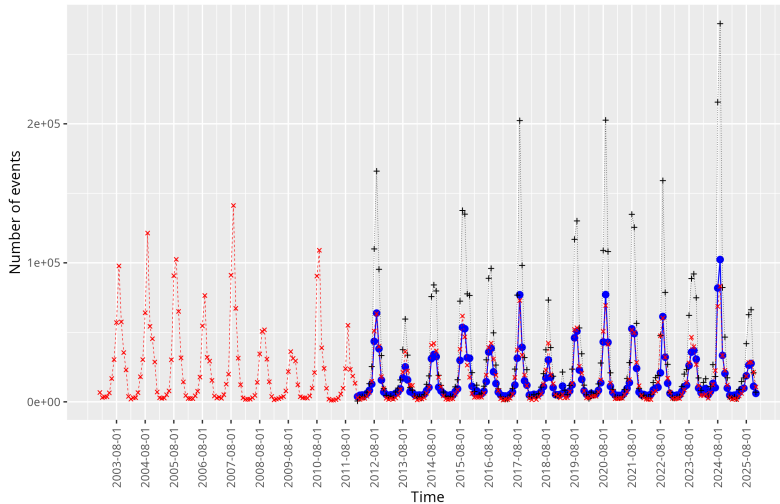
Forecast (blue line) versus observations (NPP-375D red crosses, NPP-375-PM black plusses)



# Stretch match AQUA\_M-T to match NPP-375D

Transformation of NPP-375D into AQUA\_M-T

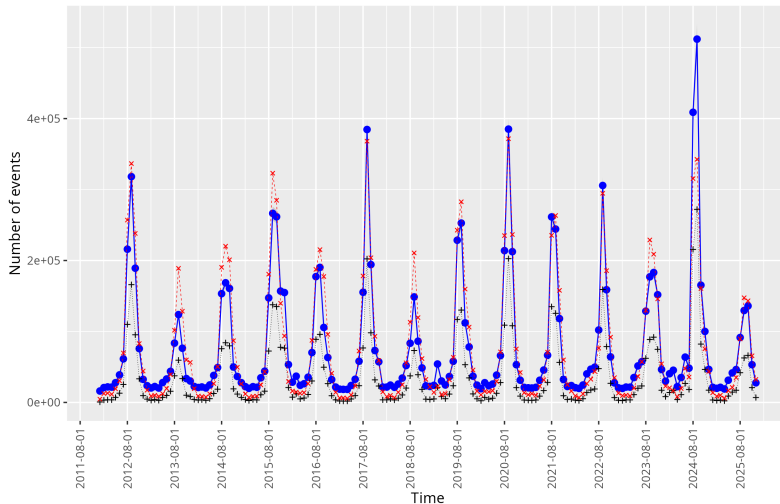
Forecast (blue line) versus observations (AQUA\_M-T red crosses, NPP-375D black plusses)



# Stretch match NPP-375-PM to match NPP-375D

Transformation of NPP-375D into NPP-375-PM

Forecast (blue line) versus observations (NPP-375-PM red crosses, NPP-375D black plusses)



*Queimadas* method (1-month regression)



## Queimadas (1-month) method for forecasting heat foci

- 1 Get time series of two reference satellites:
  - ▶ One which we would like to extend in time (A.K.A.  $y$ ).
  - ▶ The other one covering the whole time line we would like to forecast (A.K.A.  $x$ ).
- 2 Adjust a linear model ( $y = Ax + b$ ), in a month-by-month basis, to the overlapping observations in the time series of two reference satellites, using only the month of the observations.
- 3 Use the models to predict the whole time line.
- 4 Check forecast results.

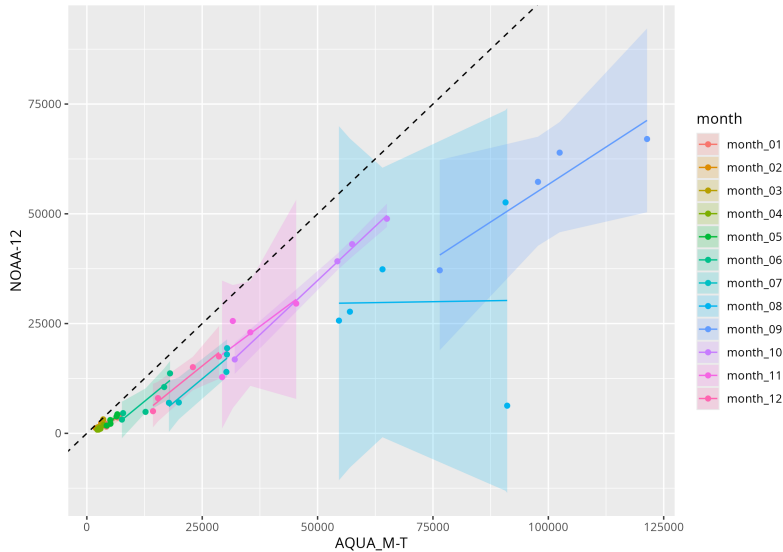


# Confidence intervals

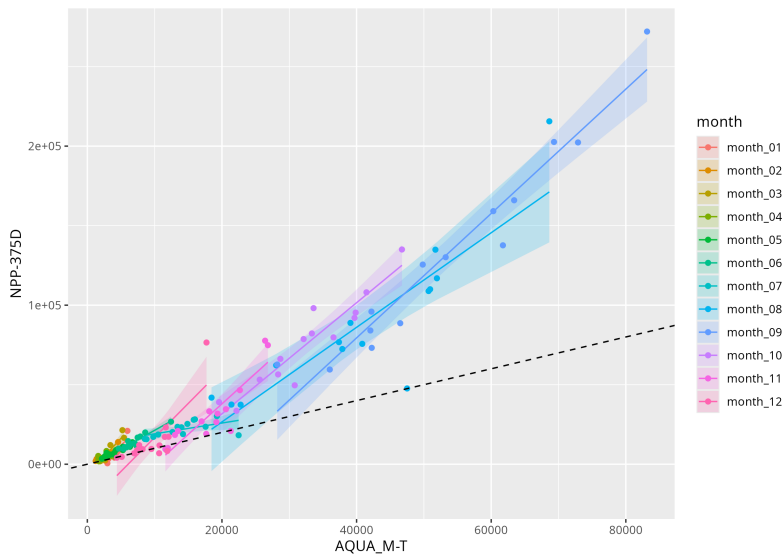
- The confidence intervals (gray shadows in the regression and forecast figures) is given at the 95% confidence level.



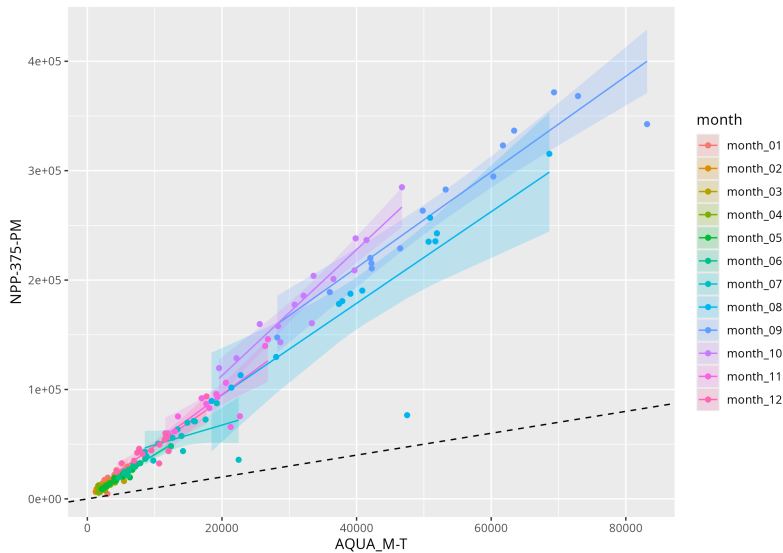
# Linear model NOAA-12 vs. AQUA\_M-T



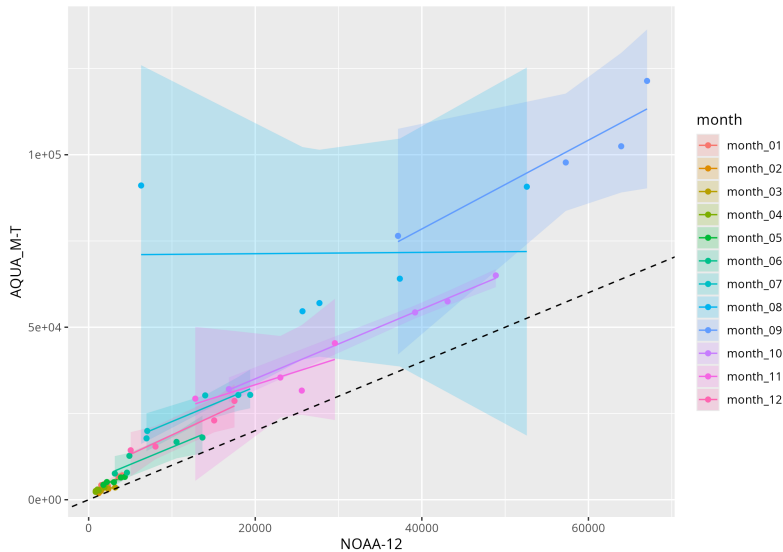
# Linear model NPP-375D vs. AQUA\_M-T



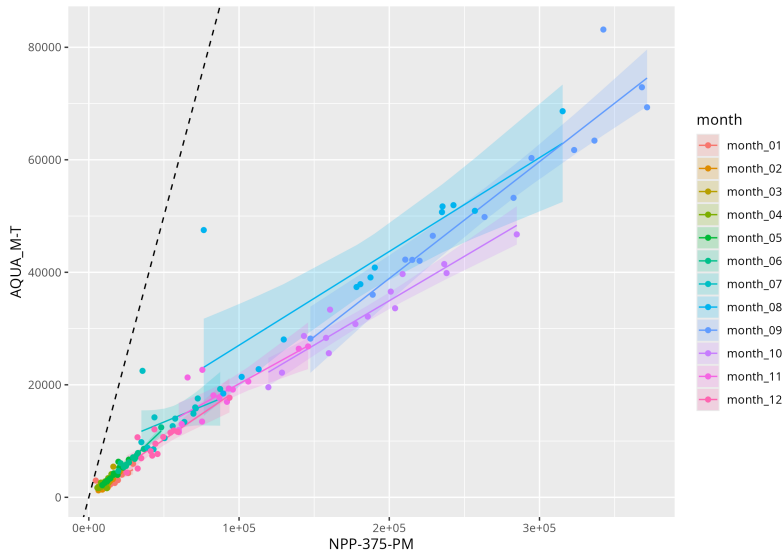
# Linear model NPP-375-PM vs. AQUA\_M-T



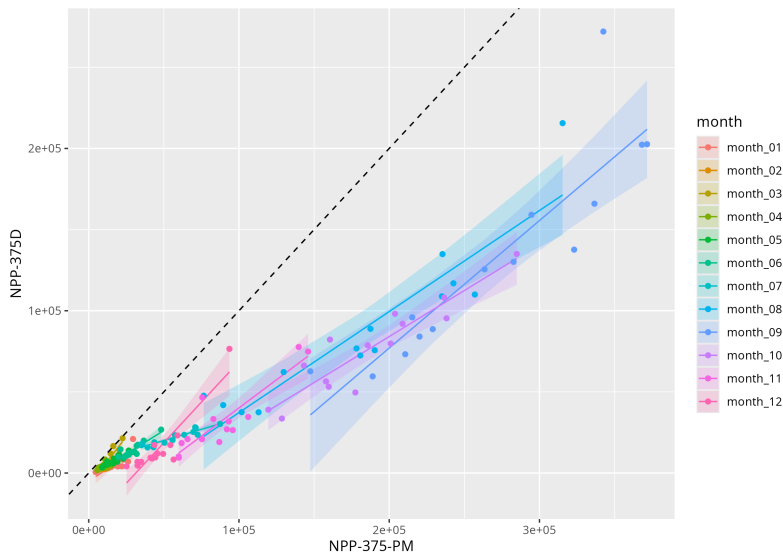
# Linear model AQUA\_M-T vs. NOAA-12



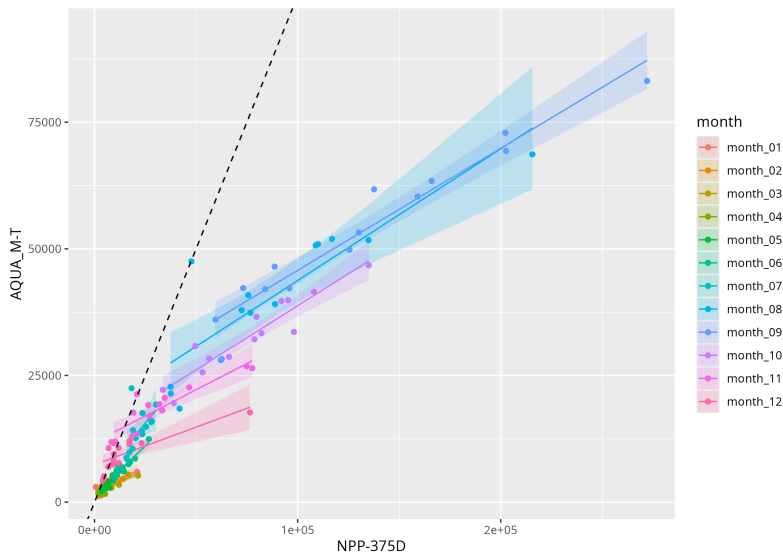
# Linear model AQUA\_M-T vs. NPP-375-PM



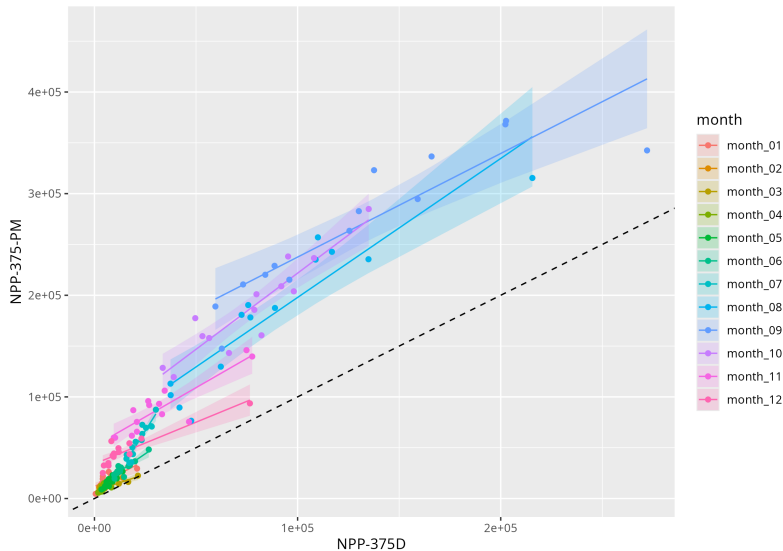
# Linear model NPP-375D vs. NPP-375-PM



# Linear model AQUA\_M-T vs. NPP-375D



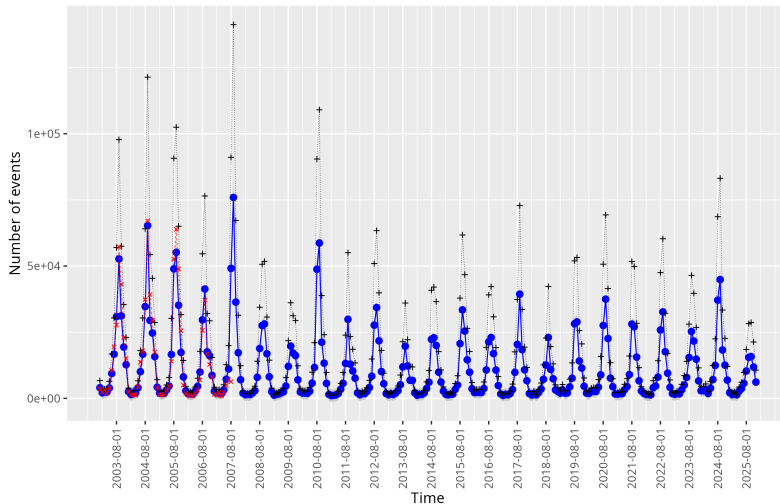
# Linear model NPP-375-PM vs. NPP-375D



# Stretch NOAA-12 to match AQUA\_M-T (month-by-month)

Transformation of AQUA\_M-T into NOAA-12

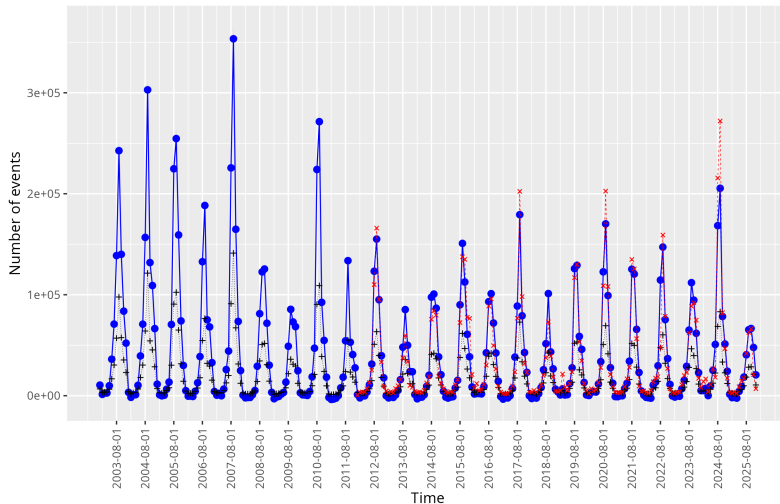
Forecast (blue line) versus observations (NOAA-12 red crosses, AQUA\_M-T black plusses)



# Stretch NPP-375D to match AQUA\_M-T (month-by-month)

Transformation of AQUA\_M-T into NPP-375D

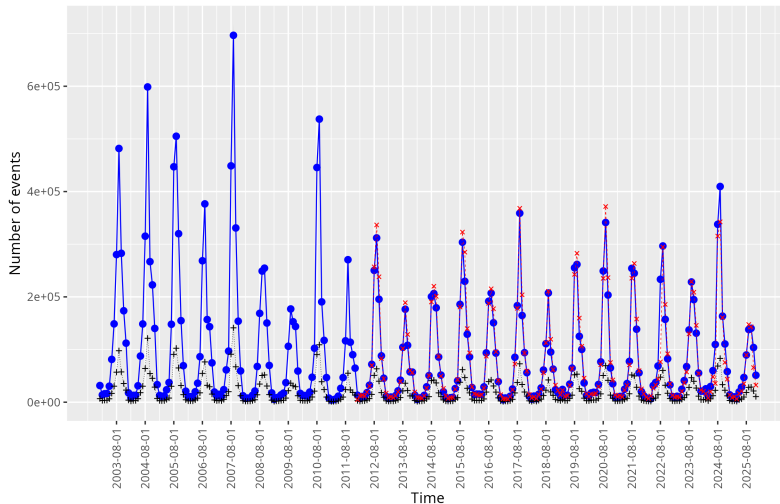
Forecast (blue line) versus observations (NPP-375D red crosses, AQUA\_M-T black plusses)



# Stretch NPP-375-PM to match AQUA\_M-T (month-by-month)

Transformation of AQUA\_M-T into NPP-375-PM

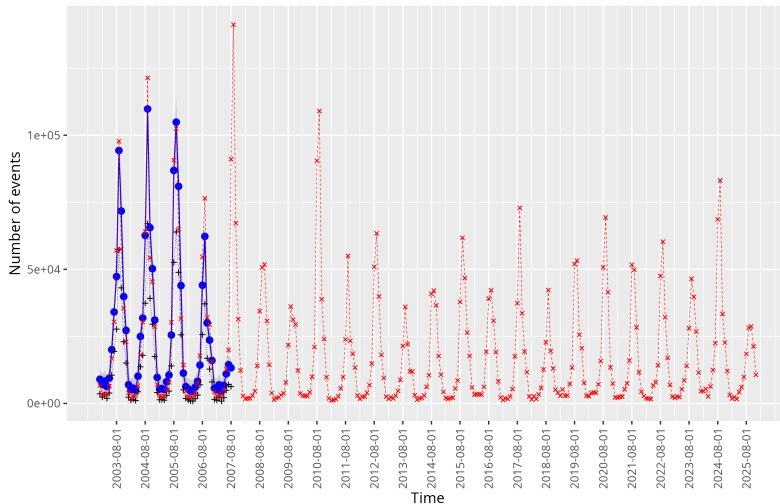
Forecast (blue line) versus observations (NPP-375-PM red crosses, AQUA\_M-T black plusses)



# Stretch match AQUA\_M-T to match NOAA-12 (month-by-month)

Transformation of NOAA-12 into AQUA\_M-T

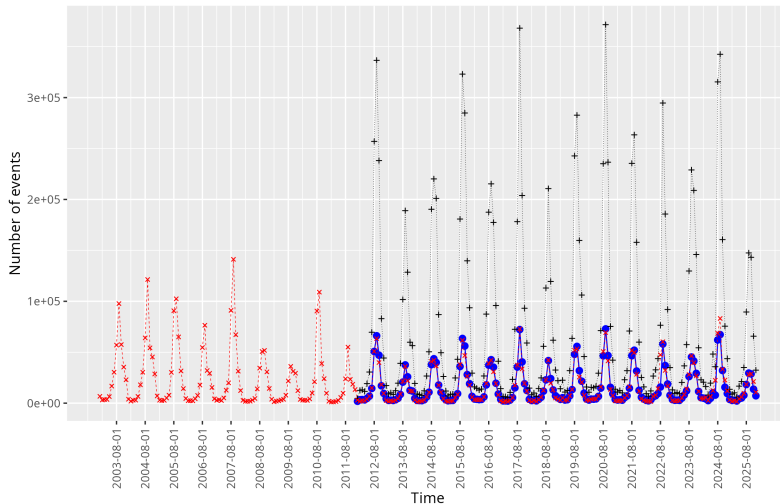
Forecast (blue line) versus observations (AQUA\_M-T red crosses, NOAA-12 black plusses)



# Stretch match AQUA\_M-T to match NPP-375-PM (month-by-month)

Transformation of NPP-375-PM into AQUA\_M-T

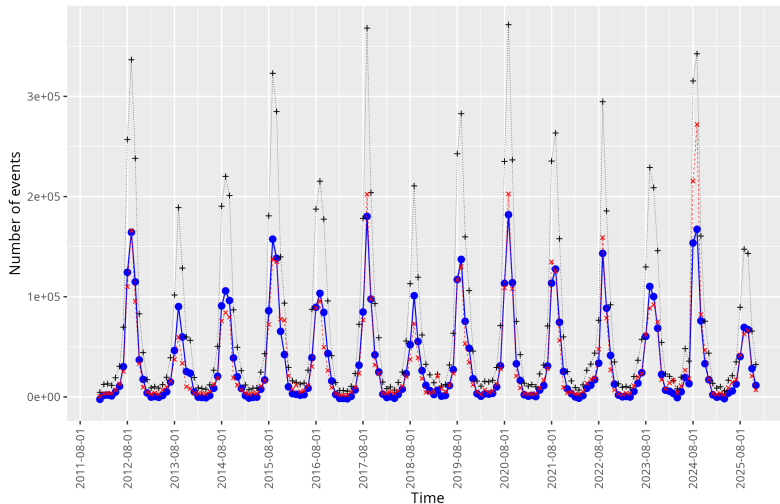
Forecast (blue line) versus observations (AQUA\_M-T red crosses, NPP-375-PM black plusses)



# Stretch match NPP-375D to match NPP-375-PM (month-by-month)

Transformation of NPP-375-PM into NPP-375D

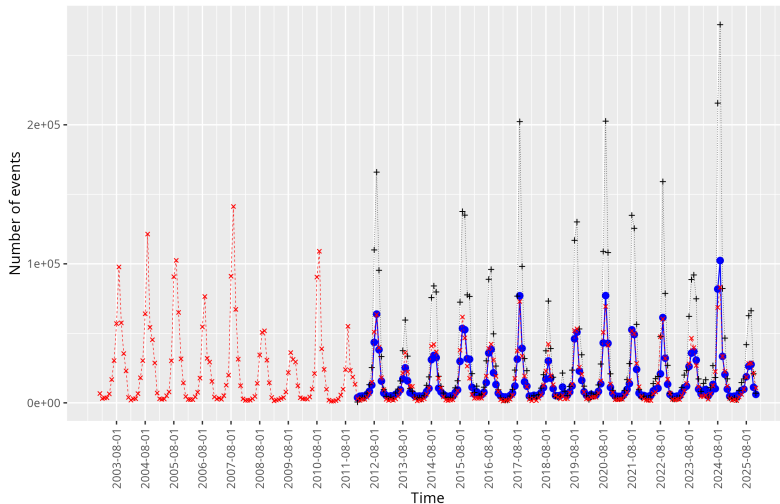
Forecast (blue line) versus observations (NPP-375D red crosses, NPP-375-PM black plusses)



# Stretch match AQUA\_M-T to match NPP-375D (month-by-month)

Transformation of NPP-375D into AQUA\_M-T

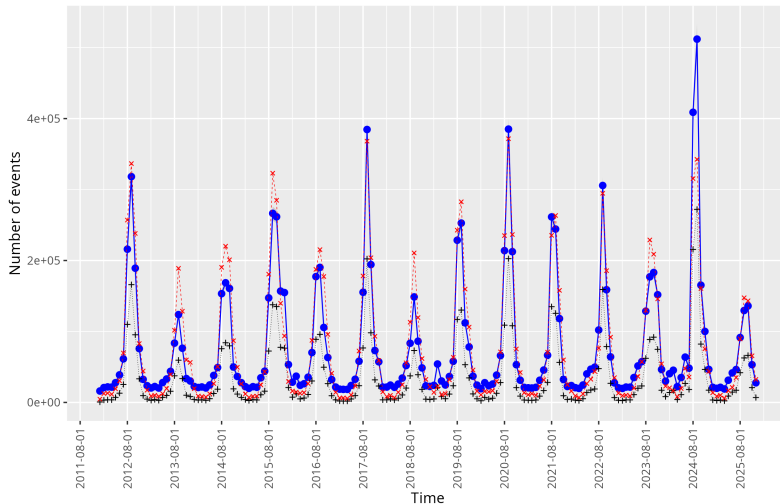
Forecast (blue line) versus observations (AQUA\_M-T red crosses, NPP-375D black plusses)



# Stretch match NPP-375-PM to match NPP-375D (month-by-month)

Transformation of NPP-375D into NPP-375-PM

Forecast (blue line) versus observations (NPP-375-PM red crosses, NPP-375D black plusses)

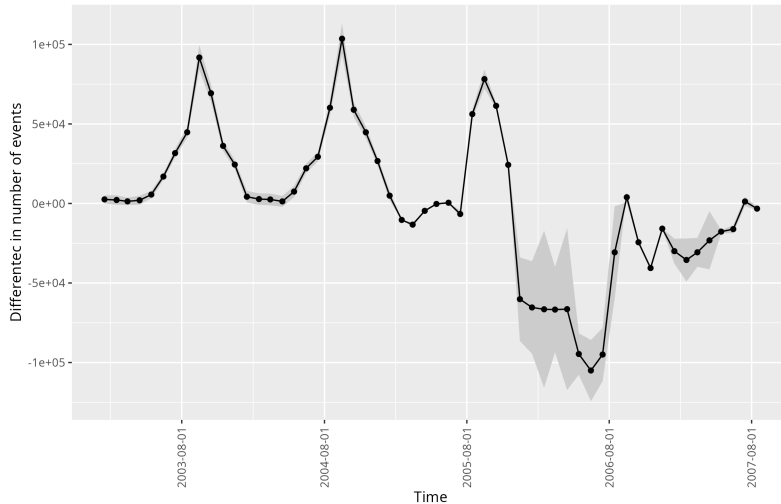


Forecast differences between Queimadas' 12 versus 1 month method

# Forecast differences NOAA-12 vs. AQUA\_M-T

Forecast differences using the Queimadas method 12 versus 1 month

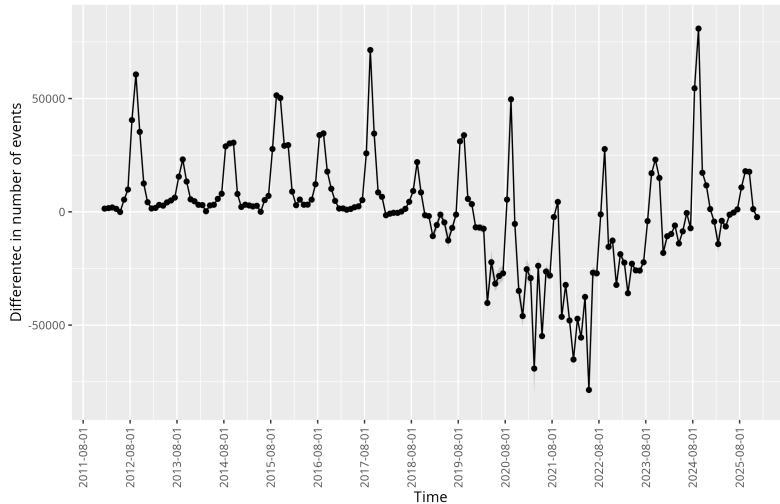
Differences in forecasts of AQUA\_M-T using NOAA-12



# Forecast differences NPP-375D vs. AQUA\_M-T

Forecast differences using the Queimadas method 12 versus 1 month

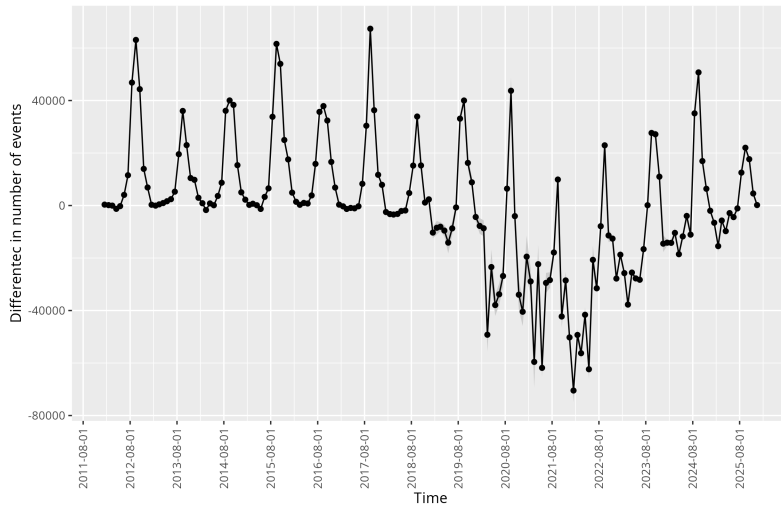
Differences in forecasts of AQUA\_M-T using NPP-375D



# Forecast differences NPP-375PM vs. AQUA\_M-T

Forecast differences using the Queimadas method 12 versus 1 month

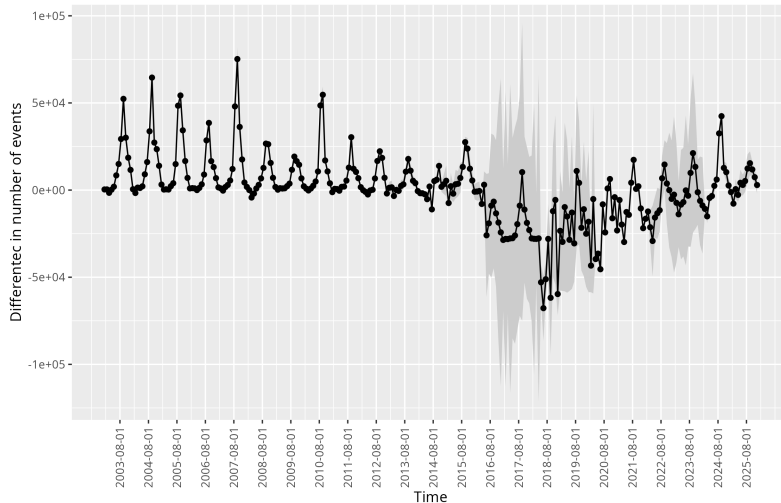
Differences in forecasts of AQUA\_M-T using NPP-375-PM



# Forecast differences AQUA\_M-T vs. NOAA-12

Forecast differences using the Queimadas method 12 versus 1 month

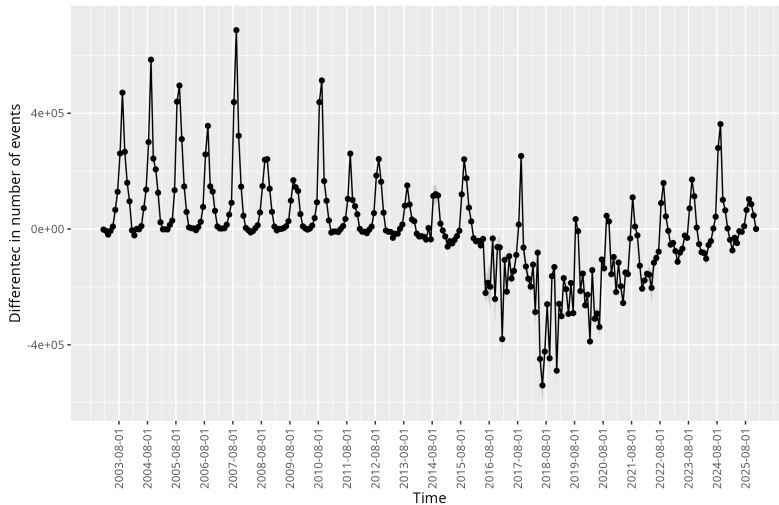
Differences in forecasts of NOAA-12 using AQUA\_M-T



# Forecast differences AQUA\_M-T vs. NPP-375PM

Forecast differences using the Queimadas method 12 versus 1 month

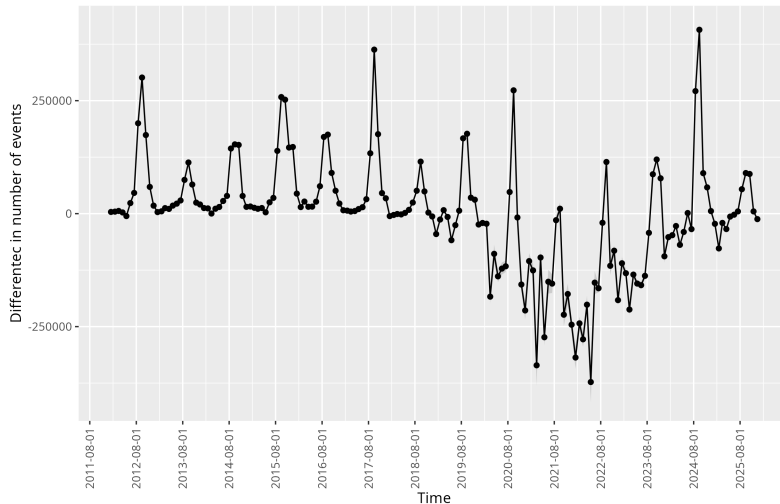
Differences in forecasts of NPP-375-PM using AQUA\_M-T



# Forecast differences NPP-375D vs. NPP-375PM

Forecast differences using the Queimadas method 12 versus 1 month

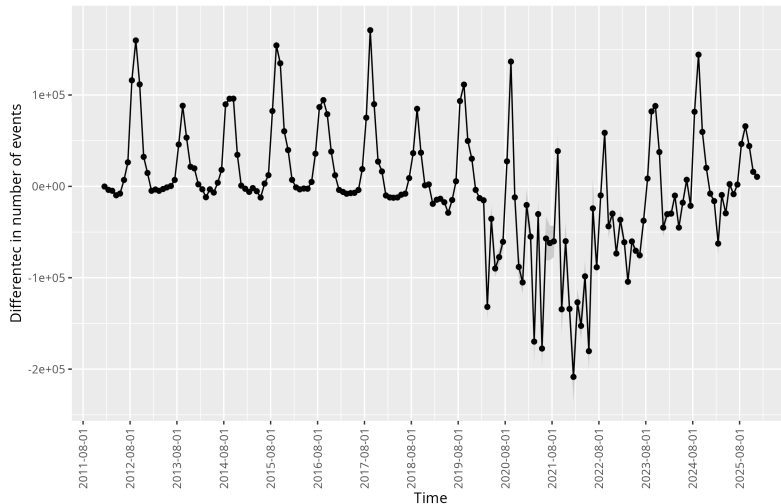
Differences in forecasts of NPP-375-PM using NPP-375D



# Forecast differences NPP-375PM vs. NPP-375D

Forecast differences using the Queimadas method 12 versus 1 month

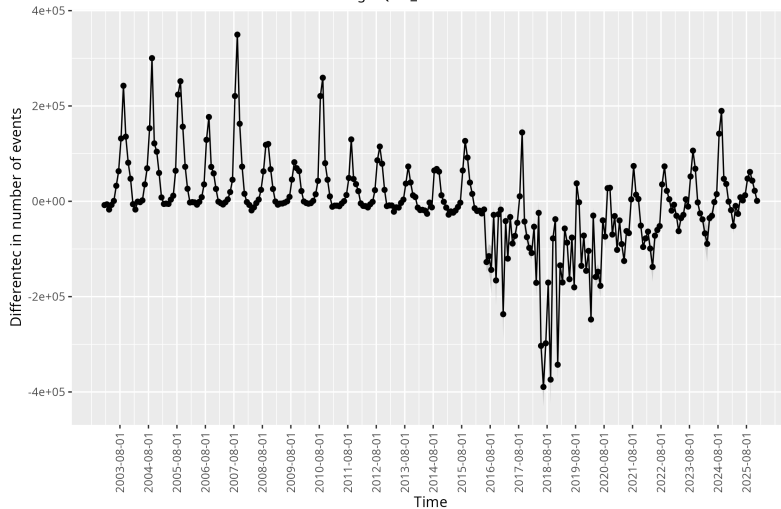
Differences in forecasts of NPP-375D using NPP-375-PM



# Forecast differences AQUA\_M-T vs. NPP-375D

Forecast differences using the Queimadas method 12 versus 1 month

Differences in forecasts of NPP-375D using AQUA\_M-T



## Future Work



## Future work

- Use a curve instead of a line for regressions. NOTE: This could have been superseded by using month-by-month regressions.
- Get data from before 2003. It's not available at Queimadas' portal:  
[https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anual/Brasil\\_todos\\_sats](https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anual/Brasil_todos_sats)
  - ▶ Use Mann-Kendall to identify breakpoints and ajust models to each segment. The *segmented* package has an implementation ??.
  - ▶ Try segmented regression on time series:
- Spatial analysis:
  - ▶ Use Landsat grid for spatial aggregation.
  - ▶ Try bfastspatial [**bfastspatial**].



## References



# References I

- [1] National Institute for Space Research in Brazil. *Banco de Dados de Queimadas*. 2023. URL: <http://www.inpe.br/queimadas/bdqueimadas> (visited on 02/11/2026).
- [2] A. C. Harvey and S. Peters. “Estimation Procedures for Structural Time Series Models”. In: *Journal of Forecasting* 9.2 (Mar. 1990), pp. 89–108. DOI: [10.1002/for.3980090203](https://doi.org/10.1002/for.3980090203).
- [3] Alberto Setzer, Fabiano Morelli, and Jean Carlos Souza. “O Banco de Dados de Queimadas Do INPE”. In: *Biodiversidade Brasileira* 9.1 (2019), pp. 239–239.
- [4] Sean J. Taylor and Benjamin Letham. “Forecasting at Scale”. In: *The American Statistician* 72.1 (Jan. 2, 2018), pp. 37–45. DOI: [10.1080/00031305.2017.1380080](https://doi.org/10.1080/00031305.2017.1380080).
- [5] Jan Verbesselt et al. “Detecting Trend and Seasonal Changes in Satellite Image Time Series”. In: *Remote Sensing of Environment* 114.1 (Jan. 2010), pp. 106–115. ISSN: 00344257. DOI: [10.1016/j.rse.2009.08.014](https://doi.org/10.1016/j.rse.2009.08.014). PMID: 16531038.



## References II

- [6] Jan Verbesselt et al. “Phenological Change Detection While Accounting for Abrupt and Gradual Trends in Satellite Image Time Series”. In: *Remote Sensing of Environment* 114.12 (Dec. 2010), pp. 2970–2980. ISSN: 00344257. DOI: [10.1016/j.rse.2010.08.003](https://doi.org/10.1016/j.rse.2010.08.003). PMID: 11818969.

